

Signals and Systems

Detailed Solutions • 1992–2026

GATE ELECTRICAL ENGINEERING

(EE)

Himanshu Agarwal & Anish Saha

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Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Signals and Systems (EE), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:





Easy



Medium



Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EE	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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SOLUTIONS

Basics of Signals

Topics

1. Basics of Signals and Mathematical Foundation
2. Step and Rectangular Signal
3. Ramp and Triangular Signal
4. Impulse Signal
5. Miscellaneous Signals
6. Complex Exponential Signal
7. Periodicity of a Signal
8. Orthogonality of Signals
9. Energy, Average and Power of a Signal
10. Even, Odd, Conjugate Symmetry Signals
11. Discrete-Time Signals and Their Types
12. Classification of Discrete-Time Signals

Unit Snapshot*

Stream: EE	**Questions:** 28	**MCQ:** 22	**MSQ:** 1	**NAT:** 5
1M: 18	**2M:** 10	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–4M(2018)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 MCQ 2026 1M Ans: D ● ○ ○

Let the sampled sequence be

$$x[n] = x_c(nT_s).$$

For the discrete-time sequence to be periodic, there must exist a positive integer N such that

$$x[n+N] = x[n].$$

Since $x_c(t)$ is periodic with period T,

$$x_c(nT_s + NT_s) = x_c(nT_s)$$

only if

$$NT_s = mT,$$

where m is a positive integer.
Hence,

$$\frac{T}{T_s} = \frac{N}{m}.$$

Since N and m are integers, $\frac{T}{T_s}$ must be a rational number.
Conversely, if

$$\frac{T}{T_s} = \frac{p}{q},$$

where p and q are integers with no common factor, then choosing

$$N = p$$

gives

$$NT_s = pT_s = qT,$$

which is an integer multiple of the continuous-time period. Therefore,

$$x[n+p] = x[n],$$

and the sequence is periodic.
Hence, $x[n]$ is periodic if and only if $\frac{T}{T_s}$ is rational.

(d) $x[n]$ will be periodic if and only if $\frac{T}{T_s}$ is a rational number.

Q1.25.1 MCQ 2025 1M Ans: A ● ○ ○

Using the sifting property of the unit impulse,

$$\int_{-\infty}^{\infty} x(t)\delta(t+3)\,dt = x(-3).$$

The given signal is

$$x(t) = -t^2\{u(t+4) - u(t-4)\}.$$

At $t = -3$,

$$u(-3+4) - u(-3-4) = u(1) - u(-7) = 1 - 0 = 1.$$

Hence,

$$x(-3) = -(-3)^2(1) = -9.$$

Therefore,

$$\int_{-\infty}^{\infty} x(t)\delta(t+3)\,dt = -9.$$

(a) -9

Key Points

- Sifting property:

$$\int_{-\infty}^{\infty} x(t)\delta(t - t_0)\, dt = x(t_0).$$

Mistakes to be avoided

- Do not evaluate the signal at $t = 3$; $\delta(t + 3)$ samples at $t = -3$.

Q1.25.2 NAT 2025 1M Ans: 7 ● ○ ○

The given signal is

$$x(t) = 1 + 2\cos(2\pi t) + 2\cos(4\pi t) + 2\cos(6\pi t).$$

The fundamental period is

$$T = 1.$$

Using orthogonality of cosine functions over one period,

$$\frac{1}{T}\int_0^T |x(t)|^2\, dt = \frac{1}{T}\int_0^T x^2(t)\, dt.$$

The DC component contributes

$$1^2 = 1.$$

Each cosine term contributes

$$\frac{(2)^2}{2} = 2.$$

Hence,

$$\frac{1}{T}\int_0^T |x(t)|^2\, dt = 1 + 2 + 2 + 2 = 7.$$

Therefore,

$$\boxed{7}$$

Key Points

- Average power of a periodic signal is

$$P = \frac{1}{T}\int_0^T |x(t)|^2\, dt.$$

- A cosine term $A\cos(\omega t)$ contributes $\frac{A^2}{2}$ to the average power.

Mistakes to be avoided

- Do not square each cosine term directly without using orthogonality.

Q1.24.1 NAT 2024 2M Ans: 2 ● ○ ○

The energy of a continuous-time signal is

$$E = \int_{-\infty}^{\infty} |x(t)|^2\, dt.$$

The transformed signal is

$$y(t) = 2x(2t - 1).$$

Using the time-shifting property, the shift by 1 does not affect the energy.

Amplitude scaling by 2 multiplies the energy by

$$2^2 = 4.$$

Time scaling by a factor of 2 changes the energy by

$$\frac{1}{2}.$$

Hence,

$$E_y = 4 \times \frac{1}{2}E = 2E.$$

Given that the energy of $2x(2t-1)$ is cE,

$$cE = 2E.$$

Therefore,

$$c = 2.$$

$$\boxed{2.0}$$

Key Points

- If $y(t) = Ax(at + b)$, then

$$E_y = \frac{|A|^2}{|a|}E.$$

- Time shifting does not change the energy of a signal.
- Amplitude scaling affects energy quadratically.

Q1.23.1 NAT | 2023 | 2M | Ans: 48 | ● ○ ○

The given discrete-time signal is

$$x[n] = 1 + 3\sin\left(\frac{15\pi}{8}n + \frac{3\pi}{4}\right) - 5\sin\left(\frac{\pi}{3}n - \frac{\pi}{4}\right).$$

The constant term is periodic for every integer N.
For a discrete-time sinusoid with frequency ω, the fundamental period is the smallest positive integer N satisfying

$$\omega N = 2\pi k, \qquad k \in \mathbb{Z}.$$

For the first sinusoid,

$$\omega_1 = \frac{15\pi}{8},$$

$$N_1 = \frac{2\pi}{15\pi/8}k = \frac{16}{15}k.$$

The smallest integer value is obtained for $k = 15$:

$$N_1 = 16.$$

For the second sinusoid,

$$\omega_2 = \frac{\pi}{3},$$

$$N_2 = \frac{2\pi}{\pi/3} = 6.$$

Hence, the fundamental period of the sum is

$$N = \text{LCM}(16, 6) = 48.$$

Therefore,

$$\boxed{48}$$

Q1.20.1 MSQ | 2020 | 1M | Ans: A,B,D | ● ○ ○

Use the orthogonality properties of trigonometric functions.
For Option A,

$$\sin(p\theta)\cos(q\theta) = \frac{1}{2}\left[\sin((p+q)\theta) + \sin((p-q)\theta)\right].$$

The average of sine over a symmetric interval is zero. Hence,

$$\lim_{a\to\infty}\frac{1}{2a}\int_{-a}^{a}\sin(p\theta)\cos(q\theta)\,d\theta = 0.$$

Therefore, **Option A is correct.**
For Option B,

$$\int_{-\pi}^{\pi}\sin(p\theta)\cos(q\theta)\,d\theta = 0$$

since $\sin(p\theta)\cos(q\theta)$ is an odd function for all real p, q. Hence, **Option B is correct.**
For Option C,

$$\sin(p\theta)\sin(q\theta) = \frac{1}{2}\left[\cos((p-q)\theta) - \cos((p+q)\theta)\right].$$

The integral over $[-\pi/2, \pi/2]$ is not zero for arbitrary real $p \neq q$. Hence, **Option C is incorrect.**
For Option D,

$$\sin(m\theta)\sin(n\theta) = \frac{1}{2}\left[\cos((m-n)\theta) - \cos((m+n)\theta)\right].$$

For non-zero integers $m \neq n$,

$$\int_{-\pi}^{\pi}\cos(k\theta)\,d\theta = 0, \qquad k \neq 0.$$

Thus,

$$\frac{1}{2\pi}\int_{-\pi}^{\pi}\sin(m\theta)\sin(n\theta)\,d\theta = 0.$$

Hence, **Option D is correct.**

A, B and D

Key Points

- Orthogonality:

$$\int_{-\pi}^{\pi}\sin(m\theta)\sin(n\theta)\,d\theta = 0, \quad m \neq n.$$

Q1.20.2 MCQ | 2020 | 1M | Ans: B | ● ○ ○

Let

$$y(t) = kx(t),$$

where k is a constant.
The average value of $x(t)$ is

$$x_A = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}x(t)\,dt.$$

Hence,

$$y_A = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}kx(t)\,dt = kx_A.$$

The RMS value of $x(t)$ is

$$x_R = \sqrt{\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}x^2(t)\,dt}.$$

Therefore,

$$y_R = \sqrt{\lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} (kx(t))^2\, dt} = \sqrt{k^2}\, x_R = |k| x_R.$$

Since the sign of k is not specified,

$$y_R \neq kx_R$$

for negative values of k.
Hence,

$$y_A = kx_A, \qquad y_R \neq kx_R.$$

(b) $y_A = kx_A;\ y_R \neq kx_R$

Q1.18.1 MCQ 2018 2M Ans: A ●○○

The given continuous-time signals are

$$x_1(t) = |t|, \qquad -1 \le t \le 1$$

$$x_2(t) = 1 - |t|, \qquad -1 \le t \le 1$$

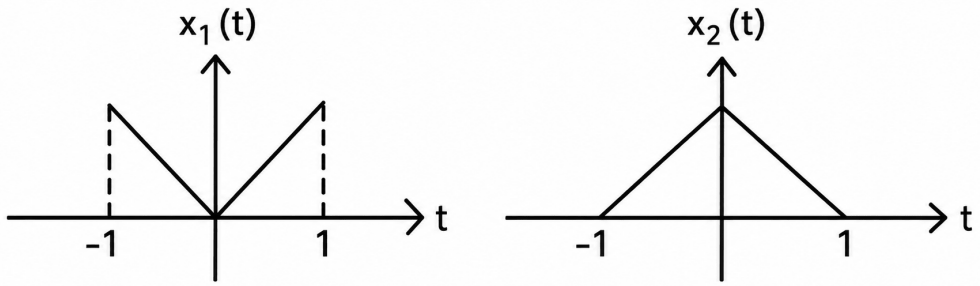


Fig. Solution Q1.18.1

Sampling with period

$$T = 0.25\ \text{s}$$

gives the samples

$$x_1[n] = \{1, 0.75, 0.5, 0.25, 0, 0.25, 0.5, 0.75, 1\}$$

$$x_2[n] = \{0, 0.25, 0.5, 0.75, 1, 0.75, 0.5, 0.25, 0\}$$

Energy of a discrete-time signal is

$$E = \sum |x[n]|^2.$$

Hence,

$$E_{x_1} = 1^2 + 0.75^2 + 0.5^2 + 0.25^2 + 0^2 + 0.25^2 + 0.5^2 + 0.75^2 + 1^2 = 3.75$$

$$E_{x_2} = 0^2 + 0.25^2 + 0.5^2 + 0.75^2 + 1^2 + 0.75^2 + 0.5^2 + 0.25^2 + 0^2 = 2.75.$$

Therefore,

$$E_{x_1} > E_{x_2}.$$

Observe that the largest-valued samples of $x_1[n]$ occur at both ends, whereas $x_2[n]$ has zeros at both ends. Since energy depends on the sum of squared samples, the two additional unit-valued samples make the energy of $x_1[n]$ greater than that of $x_2[n]$.

(a) The energy of $x_1[n]$ is greater than the energy of $x_2[n]$.

Key Points

- Energy of a discrete-time signal is $E = \sum_{n=-\infty}^{\infty} |x[n]|^2$.

Q1.18.2 MCQ 2018 2M Ans: D

Method 1

The given signal is

$$x(t) = [(t-1)u(t-1)] - [(t-2)u(t-2)] - [(t-3)u(t-3)] + [(t-4)u(t-4)].$$

Writing it in piecewise form,

$$x(t) = \begin{cases} 0, & t < 1, \\ t-1, & 1 \le t < 2, \\ 1, & 2 \le t < 3, \\ 4-t, & 3 \le t < 4, \\ 0, & t \ge 4. \end{cases}$$

The signal energy is

$$E = \int_{-\infty}^{\infty} |x(t)|^2 \, dt.$$

Hence,

$$E = \int_1^2 (t-1)^2 dt + \int_2^3 1 \, dt + \int_3^4 (4-t)^2 dt.$$

Evaluating,

$$E = \frac{1}{3} + 1 + \frac{1}{3} = \frac{5}{3}.$$

Method 2

Based on equation given:

$$x(t) = r(t-1) - r(t-2) - r(t-3) + r(t-4)$$

The graph of $x(t)$ is

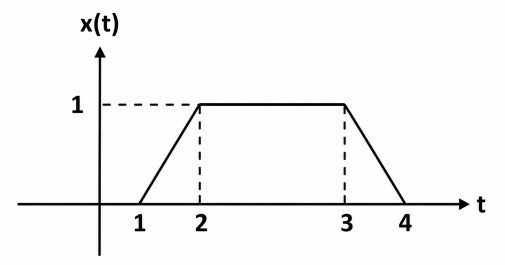


Fig. Solution Q1.18.2

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2\,dt = \int_{1}^{4} (x(t))^2\,dt$$

The area under the square of both triangles will be equal and it will be

$$\int_{1}^{2} (t-1)^2\,dt = \left[\frac{(t-1)^3}{3}\right]_1^2 = \frac{1}{3}$$

Hence,

$$E_x = 2 \times \frac{1}{3} + 1 \times 1 = \frac{5}{3} = 2.67$$

$$\boxed{\text{(d) } \frac{5}{3}}$$

Q1.17.1 NAT 2017 1M Ans: 6 ● ○ ○

From the waveform, one period is

$$T = 4.$$

Over one period,

$$f(t) = \begin{cases} 4, & \text{for 1 unit of time,} \\ -2, & \text{for 2 units of time,} \\ 0, & \text{for 1 unit of time.} \end{cases}$$

The mean square value is

$$\frac{1}{T}\int_0^T f^2(t)\,dt.$$

Hence,

$$\frac{1}{4}\left[(4)^2(1) + (-2)^2(2) + (0)^2(1)\right] = \frac{1}{4}(16+8) = \frac{24}{4} = 6.$$

$$\boxed{6}$$

Key Points

- Mean square value:

$$\overline{f^2} = \frac{1}{T}\int_0^T f^2(t)\,dt.$$

- Mean square is different from RMS. RMS = $\sqrt{\text{Mean Square}}$.

Q1.16.1 MCQ 2016 1M Ans: A ● ○ ○

Using the scaling property of the Dirac delta function,

$$\delta(2t-2) = \frac{1}{2}\delta(t-1).$$

Hence,

$$\int_{-\infty}^{\infty} e^t\,\delta(2t-2)\,dt = \frac{1}{2}\int_{-\infty}^{\infty} e^t\,\delta(t-1)\,dt.$$

Using the sifting property,

$$\int_{-\infty}^{\infty} e^{-t}\delta(t-1)\,dt = e^{-1}.$$

Therefore,

$$\int_{-\infty}^{\infty} e^{-t}\delta(2t-2)\,dt = \frac{1}{2e}.$$

$$\boxed{\text{(a) } \frac{1}{2e}}.$$

Key Points

- Scaling property:

$$\delta(at-b) = \frac{1}{|a|}\delta\left(t-\frac{b}{a}\right).$$

- Sifting property:

$$\int_{-\infty}^{\infty} x(t)\delta(t-t_0)\,dt = x(t_0).$$

Q1.16.2 MCQ 2016 1M Ans: D ●○○

Using the standard Dirichlet integral,

$$\int_{-\infty}^{\infty} \frac{\sin(at)}{\pi t}\,dt = \begin{cases} 1, & a > 0, \\ -1, & a < 0. \end{cases}$$

Here,

$$a = 2 > 0.$$

Therefore,

$$\int_{-\infty}^{\infty} \frac{\sin(2\pi t)}{\pi t}\,dt = 1.$$

Hence,

$$2\int_{-\infty}^{\infty} \frac{\sin(2\pi t)}{\pi t}\,dt = 2 \times 1 = 2.$$

$$\boxed{\text{(d) } 2}$$

Q1.15.1 NAT 2015 1M Ans: 90 ●○○

The moving average system is

$$y(t) = \frac{1}{T}\int_{t-T}^{t} u(\tau)\,d\tau.$$

This is an LTI system with impulse response

$$h(t) = \begin{cases} \frac{1}{T}, & 0 \le t \le T, \\ 0, & \text{otherwise.} \end{cases}$$

Its frequency response is

$$H(j\omega) = \frac{1}{T}\int_{0}^{T} e^{-j\omega t}\,dt = e^{-j\omega T/2}\frac{\sin(\omega T/2)}{\omega T/2}.$$

The input frequency is

$$f = \frac{1}{2T}\ \text{Hz},$$

so

$$\omega = 2\pi f = \frac{\pi}{T}.$$

Hence,

$$\angle H(j\omega) = -\frac{\omega T}{2} = -\frac{\pi}{2} = -90^\circ.$$

Since

$$\frac{\sin(\pi/2)}{\pi/2} > 0,$$

there is no additional phase shift.
Therefore, the output lags the input by

$$\boxed{90}$$

Q1.14.1 MCQ 2014 2M Ans: A ●○○

The waveform consists of three parts:

$$x(t) = \begin{cases} 1, & 0 \le t < T, \\ \dfrac{t-T}{T}, & T \le t < 2T, \\ 1, & t \ge 2T. \end{cases}$$

The first part is represented by

$$u(t) - u(t-T).$$

The ramp between T and $2T$ is

$$\frac{t-T}{T}u(t-T) - \frac{t-2T}{T}u(t-2T),$$

which starts at 0 when $t = T$ and becomes 1 at $t = 2T$.
For $t \ge 2T$, the above ramp remains equal to 1, matching the waveform.
Hence,

$$x(t) = u(t) - u(t-T) + \frac{t-T}{T}u(t-T) - \frac{t-2T}{T}u(t-2T).$$

$$\boxed{\text{(a) } u(t) - u(t-T) + \frac{(t-T)}{T}u(t-T) - \frac{(t-2T)}{T}u(t-2T)}$$

Key Points

- Represent piecewise signals by adding delayed unit-step and ramp functions.
- A finite-duration ramp is obtained by subtracting two shifted ramps.

Q1.11.1 MCQ 2011 2M Ans: A ●○○

Given that the random signal has zero mean,

$$\mu = 0.$$

The mean square value is

$$E[X^2] = \sigma^2.$$

For a uniform random variable distributed over the interval $[-a, a]$,

$$\sigma^2 = \frac{a^2}{3}.$$

The RMS value is

$$X_{\text{RMS}} = \sqrt{E[X^2]} = \sqrt{\sigma^2} = \sqrt{\frac{a^2}{3}} = \frac{a}{\sqrt{3}}.$$

$$\boxed{\text{(a) } \frac{a}{\sqrt{3}}}$$

Q1.10.1 MCQ | 2010 | 1M | Ans: D | ● ○ ○

The given signal is

$$x(t) = 8\sin\left(0.8\pi t + \frac{\pi}{4}\right).$$

Comparing with the standard form

$$x(t) = A\sin(\omega t + \phi),$$

the angular frequency is

$$\omega = 0.8\pi \text{ rad/s}.$$

The time period is

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{0.8\pi} = \frac{2}{0.8} = 2.5 \text{ s}.$$

The amplitude and phase shift do not affect the period.

$$\boxed{\text{(d) } 2.5 \text{ s}}$$

Q1.08.1 MCQ | 2008 | 2M | Ans: B | ● ○ ○

Advancing the input by 3 samples gives $z_1[n] = x[n+3]$. Downsampling that sequence by 4 gives $z_2[n] = x[4n+3]$. Time reversal then gives

$$z_2[-n] = x[3-4n],$$

which is the required signal. Thus option $\boxed{\text{B}}$ is correct.
For comparison, (A) gives $x[-4n-3]$, (C) gives $x[-4n-12]$, and (D) gives $x[12-4n]$, so none matches.

Q1.06.1 MCQ | 2006 | 1M | Ans: MTA | ● ○ ○

Warning for answer checking: The word "finite" is not defined in the question. Under the pointwise-finite definition, (A) is false and (D) is true.
Pointwise finiteness means $|x(t)| < \infty$ at each finite time. Boundedness is stronger: there must be one finite constant M such that $|x(t)| \le M$ at every time. For example, $x(t) = t$ is finite at each finite t but has no uniform bound. Thus pointwise finite does not imply bounded.
Conversely, a bounded signal is pointwise finite. This makes (D) true under that meaning of "finite." A bounded signal need not have finite energy or duration: $x(t) = 1$ is a counterexample to (B) and (C).
If "finite" was intended to mean finite duration or some other signal class, that definition must be verified from the original source. Finite duration by itself also does not guarantee bounded amplitude without further regularity assumptions.
Marks to all: the term "finite is undefined here. Under the usual meaning (finite at every instant), D is true.

Q1.05.1 MCQ 2005 2M Ans: A ● ○ ○

For one time period, the waveform is

$$V(t) = \begin{cases} \frac{2t}{T}, & 0 \le t \le \frac{T}{2} \\ 0, & \frac{T}{2} < t \le T \end{cases}$$

The RMS value is

$$V_{\text{RMS}} = \sqrt{\frac{1}{T}\int_0^T V^2(t)\,dt} = \sqrt{\frac{1}{T}\int_0^{T/2}\left(\frac{2t}{T}\right)^2 dt}.$$

Evaluating the integral,

$$V_{\text{RMS}} = \sqrt{\frac{4}{T^3}\left[\frac{t^3}{3}\right]_0^{T/2}} = \sqrt{\frac{4}{T^3}\cdot\frac{T^3}{24}} = \sqrt{\frac{1}{6}}.$$

$$\boxed{\text{(a) } \sqrt{\frac{1}{6}}}$$

Q1.04.1 MCQ 2004 2M Ans: A ● ○ ○

Over one time period,

$$i(t) = \begin{cases} -\frac{12t}{T}, & 0 \le t \le \frac{T}{2}, \\ 6, & \frac{T}{2} < t \le T. \end{cases}$$

The RMS value is

$$I_{\text{RMS}} = \sqrt{\frac{1}{T}\int_0^T i^2(t)\,dt}.$$

Substituting the waveform,

$$I_{\text{RMS}} = \sqrt{\frac{1}{T}\left[\int_0^{T/2}\left(\frac{12t}{T}\right)^2 dt + \int_{T/2}^T 6^2\,dt\right]}.$$

$$I_{\text{RMS}} = \sqrt{\frac{1}{T}\left[\frac{144}{T^2}\cdot\frac{(T/2)^3}{3} + 36\cdot\frac{T}{2}\right]} = \sqrt{6+18} = \sqrt{24} = 2\sqrt{6}\,\text{A}.$$

$$\boxed{\text{Option A: } 2\sqrt{6}\,\text{A}}$$

Q1.04.2 MCQ 2004 1M Ans: B ● ○ ○

The resultant current is

$$i(t) = I_{DC} + i_{AC}(t),$$

where

$$I_{DC} = 10\,\text{A}, \qquad I_{AC(\text{peak})} = 20\,\text{A}.$$

The RMS value of the AC component is

$$I_{AC(\text{RMS})} = \frac{20}{\sqrt{2}} = 10\sqrt{2}\,\text{A}.$$

Since the average value of the AC component over one cycle is zero,

$$I_{\text{RMS}} = \sqrt{I_{DC}^2 + I_{AC(\text{RMS})}^2}.$$

Substituting the values,

$$I_{\text{RMS}} = \sqrt{10^2 + \left(10\sqrt{2}\right)^2} = \sqrt{100 + 200} = \sqrt{300} = 10\sqrt{3} \approx 17.3\,\text{A}.$$

(b) 17.3 A

Q1.02.1 MCQ 2002 2M Ans: D ●○○

The waveform alternates between +100 V and −100 V with equal magnitude. Since RMS depends on the square of the voltage, the sign does not affect the result.

$$V_{\text{RMS}} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt}.$$

Here,

$$v^2(t) = 100^2 \quad \text{for the entire period.}$$

Hence,

$$V_{\text{RMS}} = \sqrt{\frac{1}{T}\int_0^T 100^2\,dt} = \sqrt{10000} = 100\,\text{V}.$$

(d) 100 V

Q1.02.2 MCQ 2002 1M Ans: D ●○○

An arbitrary number of harmonics changes the shape of the waveform. Hence, the waveform parameters such as peak value, RMS value, and average value are generally different from those of the fundamental component alone.
Therefore, the overall waveform differs from its fundamental component in all three measures.

(d) All the three measures (peak, RMS and average values)

Q1.02.3 MCQ 2002 1M Ans: D ●○○

For a capacitor,

$$i(t) = C\frac{dv_c(t)}{dt}.$$

Given

$$i(t) = 5\delta(t).$$

Hence,

$$\frac{dv_c(t)}{dt} = \frac{5}{C}\delta(t).$$

Integrating both sides,

$$v_c(t) = \frac{5}{C}\int \delta(t)\,dt = \frac{5}{C}u(t),$$

assuming zero initial voltage across the capacitor.

$$\boxed{\text{(d) } v_c(t) = \frac{5u(t)}{C}}$$

Q1.99.1 MCQ | 1999 | 1M | Ans: A | ● ○ ○

For a half-wave rectified symmetrical square wave,

$$i(t) = \begin{cases} 2\,\text{A}, & 0 \le t < \dfrac{T}{2}, \\ 0, & \dfrac{T}{2} \le t < T. \end{cases}$$

The RMS value is

$$I_{\text{RMS}} = \sqrt{\frac{1}{T}\int_0^T i^2(t)\,dt}.$$

Substituting the waveform,

$$I_{\text{RMS}} = \sqrt{\frac{1}{T}\left[\int_0^{T/2} (2)^2\,dt\right]} = \sqrt{\frac{4(T/2)}{T}} = \sqrt{2}\,\text{A}.$$

$$\boxed{\text{(a) } \sqrt{2}\,\text{A}}$$

Q1.95.1 MCQ | 1995 | 1M | Ans: B | ● ○ ○

Over one time period,

$$s(t) = \begin{cases} \dfrac{2A}{T}t, & 0 \le t \le \dfrac{T}{2}, \\ -A, & \dfrac{T}{2} < t \le T. \end{cases}$$

The RMS value is

$$S_{\text{RMS}} = \sqrt{\frac{1}{T}\int_0^T s^2(t)\,dt}.$$

Substituting the waveform,

$$S_{\text{RMS}} = \sqrt{\frac{1}{T}\left[\int_0^{T/2}\left(\frac{2At}{T}\right)^2 dt + \int_{T/2}^{T} A^2\,dt\right]}.$$

Evaluating the integrals,

$$S_{\text{RMS}} = \sqrt{\frac{1}{T}\left[\frac{A^2T}{6} + \frac{A^2T}{2}\right]} = \sqrt{\frac{2A^2}{3}} = A\sqrt{\frac{2}{3}}.$$

$$\boxed{\text{(b) } \sqrt{\frac{2}{3}}A}$$

Q1.94.1 MCQ | 1994 | 1M | Ans: B | ● ○ ○

Using the sampling property of the Dirac delta function,

$$\int_{-\infty}^{\infty} f(t)\,\delta(t-a)\,dt = f(a),$$

provided a lies within the limits of integration.
Here,

$$f(t) = e^{-2t}, \qquad a = 1.$$

Since $1 \in [-5, 6]$,

$$\int_{-5}^{6} e^{-2t}\delta(t-1)\,dt = e^{-2(1)} = e^{-2}.$$

$$\boxed{\text{(b) } e^{-2}}$$

Q1.94.2 MCQ | 1994 | 1M | Ans: B | ● ○ ○

Match each waveform with its mathematical expression.

(A) : Ramp starting from the origin $\Rightarrow f(t) = t\,u(t)$ (R)
(B) : Ramp delayed by 1 $\Rightarrow f(t) = (t-1)u(t-1)$ (U)
(C) : Ramp with intercept 1 $\Rightarrow f(t) = (t+1)u(t)$ (S)
(D) : Delayed ramp shifted upward by 1 $\Rightarrow f(t) = t\,u(t-1)$ (P)

$$\boxed{\text{(b) } A-R,\ B-U,\ C-S,\ D-P}$$

SOLUTIONS

11

Basics of Systems

Topics

1. Continuous-Time Convolution
2. Discrete-Time Convolution
3. Linear and Non-Linear Systems
4. Time Variant and Time-Invariant Systems
5. LTI Systems - Definition and Properties
6. Causal and Non-Causal Systems
7. Static and Dynamic Systems
8. Stable and Unstable Systems
9. Invertible and Non-Invertible Systems
10. Systems Involving Differential Equations

Unit Snapshot*

Stream: EE	**Questions:** 44	**MCQ:** 40	**MSQ:** 1	**NAT:** 3
1M: 21	**2M:** 22	**Descriptive:** 1	**Avg Weightage** ~3M	**Range:** 0M(2014)–6M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q2.25.1 MCQ 2025 2M Ans: B ●○○

The impulse response of the given continuous-time LTI system is

$$h(t) = u(t)$$

since the output corresponding to the unit impulse input is the unit step signal.
For a continuous-time LTI system, BIBO stability requires

$$\int_{-\infty}^{\infty} |h(t)|\, dt < \infty.$$

Here,

$$\int_{-\infty}^{\infty} |u(t)|\, dt = \int_{0}^{\infty} 1\, dt = \infty.$$

Hence, the system is **not BIBO stable**. Therefore, there exists at least one bounded input that produces an unbounded output. Since the system is not BIBO stable, it violates the definition of bounded-input bounded-output stability. Thus, a bounded input can generate an unbounded output, making only option B correct.

(b) It is possible to find a bounded input signal which when applied to S results in an unbounded output signal.

Key Points

- A continuous-time LTI system is BIBO stable if and only if

$$\int_{-\infty}^{\infty} |h(t)|\, dt < \infty.$$

- If this condition is not satisfied, at least one bounded input will produce an unbounded output.

Mistakes to be avoided

- Remember that BIBO stability depends on the *absolute integrability* of the impulse response, not on its boundedness.

Q2.25.2 MCQ 2025 2M Ans: A ●○○

Using the convolution property,

$$\int_{-\infty}^{\infty} y(t)\, dt = \left(\int_{-\infty}^{\infty} x_1(t)\, dt\right)\left(\int_{-\infty}^{\infty} x_2(t)\, dt\right).$$

Now,

$$\int_{-\infty}^{\infty} x_1(t)\, dt = \int_{0}^{1} 1\, dt + \int_{1}^{2} (2-t)\, dt = 1 + \frac{1}{2} = \frac{3}{2}.$$

Similarly,

$$\int_{-\infty}^{\infty} x_2(t)\, dt = \int_{0}^{1} t\, dt + \int_{1}^{2} (2-t)\, dt = \frac{1}{2} + \frac{1}{2} = 1.$$

Hence,

$$\int_{-\infty}^{\infty} y(t)\, dt = \frac{3}{2} \times 1 = \frac{3}{2} = 1.5.$$

Instead of evaluating the convolution explicitly, use the area property of convolution:

$$\text{Area of } (x_1 * x_2) = (\text{Area of } x_1)(\text{Area of } x_2).$$

Computing the areas of the two given signals directly gives

$$\frac{3}{2} \times 1 = 1.5.$$

(a) 1.5

Q2.25.3 MCQ 2025 1M Ans: B ● ○ ○

A stable first-order LTI system has the unit step response

$$y(t) = K\left(1 - e^{-t/\tau}\right),$$

where K is the steady-state value and τ is the time constant.
From the table,

$$K = y(\infty) = 3.$$

At one time constant,

$$y(\tau) = 0.632\,K = 0.632 \times 3 = 1.896.$$

From the given data,

$$y(1.6) = 1.65, \qquad y(2.6) = 2.18.$$

Since 1.896 lies much closer to the output at $t = 2.6\,\text{s}$, the time constant is approximately

$$\tau \approx 2\,\text{s}.$$

Using

$$3\left(1 - e^{-t/\tau}\right) = 2.18,$$

we get

$$e^{-2.6/\tau} = 1 - \frac{2.18}{3} = 0.2733.$$

Hence,

$$\tau = \frac{2.6}{-\ln(0.2733)} \approx 1.99\,\text{s}.$$

Thus,

$$\tau \approx 2\,\text{s}.$$

(b) 2

Q2.25.4 MCQ 2025 1M Ans: A ● ○ ○

For a discrete-time LTI system, the output is given by the convolution of the input with the impulse response:

$$y[n] = x[n] * h[n],$$

where

$$h[n] = S\{\delta[n]\} = \begin{cases} 1, & n \in \{0, 1, 2\}, \\ 0, & \text{otherwise.} \end{cases}$$

Method 1

The impulse response can be written as

$$h[n] = \delta[n] + \delta[n-1] + \delta[n-2].$$

Hence,

$$\begin{aligned} y[n] &= x[n] * h[n] \\ &= x[n] * (\delta[n] + \delta[n-1] + \delta[n-2]) \\ &= x[n] + x[n-1] + x[n-2]. \end{aligned}$$

Therefore, the correct expression matches Option A.

Method 2

Using the convolution sum,

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k].$$

Since $h[n-k]$ is non-zero only for

$$n - k = 0, 1, 2,$$

we have

$$k = n,\ n-1,\ n-2.$$

Thus,

$$y[n] = x[n] + x[n-1] + x[n-2].$$

$$\boxed{\text{(a) } x[n] + x[n-1] + x[n-2]}$$

Q2.24.1 MCQ | 2024 | 2M | Ans: D | ● ○ ○

Method 1

Assume the input is bounded:

$$|x[n]| \le M, \qquad \forall n.$$

For system T_1,

$$(T_1 x)[n] = \sum_{k=0}^{n} x[k].$$

Hence,

$$|(T_1 x)[n]| \le \sum_{k=0}^{n} |x[k]| \le (n+1)M.$$

Since the bound grows with n, the output can become unbounded even for a bounded input. Therefore,

T_1 is not BIBO stable.

For system T_2,

$$(T_2 x)[n] = \sum_{k=0}^{n} \frac{1}{2^k} x[k].$$

Thus,

$$\begin{aligned} |(T_2 x)[n]| &\le M \sum_{k=0}^{n} \frac{1}{2^k} \\ &\le M \sum_{k=0}^{\infty} \frac{1}{2^k} \\ &= M \left(\frac{1}{1 - \frac{1}{2}} \right) = 2M. \end{aligned}$$

Since the output remains bounded for every bounded input,

T_2 is BIBO stable.

Method 2

Choose the bounded input

$$x[n] = 1, \qquad \forall n.$$

Then,

$$(T_1 x)[n] = n + 1,$$

which is unbounded.
Also,

$$(T_2 x)[n] = 1 + \frac{1}{2} + \cdots + \frac{1}{2^n} = 2 - \frac{1}{2^n},$$

which is bounded by 2.
Hence, T_1 is not BIBO stable, whereas T_2 is BIBO stable.

(d) T_1 is not BIBO stable but T_2 is BIBO stable.

Q2.24.2 MCQ 2024 2M Ans: B ● ○ ○

Combine the exponential factors inside the integral:

$$y(t) = \int_{-\infty}^{t} e^{-(t-\tau)} x(\tau)\, d\tau.$$

This is convolution with the fixed impulse response

$$h(t) = e^{-t} u(t).$$

Convolution with a fixed impulse response is linear and time-invariant, so option B is correct. The response is also causal because $h(t) = 0$ for $t < 0$.
The outside factor e^{-t} does not establish time variance by itself: its combination with e^{τ} yields a kernel depending only on $t - \tau$. That difference dependence is what matters.

Q2.24.3 MCQ 2024 1M Ans: A ● ○ ○

Let

$$y(t) = x(-t).$$

The convolution of $x(t)$ and $y(t)$ is

$$z(t) = x(t) * y(t) = \int_{-\infty}^{\infty} x(\tau)\, y(t-\tau)\, d\tau.$$

Method 1

Substituting $y(t) = x(-t)$,

$$z(t) = \int_{-\infty}^{\infty} x(\tau) x(\tau - t)\, d\tau.$$

Now,

$$z(-t) = \int_{-\infty}^{\infty} x(\tau) x(\tau + t)\, d\tau.$$

Let

$$\lambda = \tau + t.$$

Then,

$$\begin{aligned} z(-t) &= \int_{-\infty}^{\infty} x(\lambda - t) x(\lambda)\, d\lambda \\ &= \int_{-\infty}^{\infty} x(\lambda) x(\lambda - t)\, d\lambda \\ &= z(t). \end{aligned}$$

Hence,

$$z(-t) = z(t),$$

so the convolution is an even signal.

Method 2

The convolution

$$x(t) * x(-t)$$

is the autocorrelation of $x(t)$. A fundamental property of the autocorrelation function is that it is always an even function.

(a) It is an even signal.

Q2.23.1 MSQ | 2023 | 1M | Ans: B

Check each statement individually.
Option A: False.
Causality and stability are independent properties. A causal LTI system need not be BIBO stable.
Option B: True.
For a discrete-time LTI system,

$$\text{System is causal} \iff h[n] = 0, \quad n < 0,$$

where $h[n]$ is the response to the unit impulse input.
Option C: False
For stability,

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty.$$

Consider a signal

$$h[n] = \tan\left(\frac{n\pi}{4}\right), \qquad 0 < n < 4.$$

which is of finite duration. Here, at $n = 2$,

$$\tan\left(\frac{\pi}{2}\right)$$

it is undefined (tends to ∞), which does not satisfy condition for BIBO stable system.
Option D: False.
The condition

$$0 < |h[n]| < 1, \qquad \forall n$$

does *not* guarantee BIBO stability. For stability, the impulse response must satisfy

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty.$$

A counterexample is

$$h[n] = \begin{cases} \dfrac{1}{n+1}, & n \geq 0, \\ 0, & n < 0. \end{cases}$$

Clearly,

$$0 < |h[n]| < 1, \qquad \forall n \geq 0,$$

but

$$\sum_{n=-\infty}^{\infty} |h[n]| = \sum_{n=0}^{\infty} \frac{1}{n+1} = \infty.$$

Since the impulse response is *not* absolutely summable, the system is **not** BIBO stable.
Hence, Option D is incorrect.

(b) A discrete-time LTI system is causal if and only if its response to a step input $u[n]$ is 0 for $n < 0$.

Key Points

- The statement says that the impulse response is of *finite duration*. This means that $h[n]$ is non-zero only over a finite number of samples *and is a valid finite-valued sequence*. Hence,

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty,$$

so the system is always BIBO stable.

But if we consider examples such as

$$h[n] = \tan\left(\frac{n\pi}{4}\right), \qquad 0 < n < 4.$$

Here, at $n = 2$,

$$\tan\left(\frac{\pi}{2}\right)$$

is undefined (tends to ∞). Such a sequence is *not a valid finite-valued impulse response* of an LTI system. But GATE official answer key mentions only Option B to be true, so that means they have considered the word 'finite valued' to be explicit which was not mentioned in question statement

Hence, whenever GATE states that an impulse response has *finite duration*, it is implicitly assumed to be a well-defined finite-valued sequence. Therefore, Option C is not always **true**.

Q2.23.2 NAT 2023 1M Ans: 4 ●○○

The signals are

$$x(t) = \begin{cases} 1, & -1 \le t \le 1, \\ 0, & \text{otherwise}, \end{cases}$$

and

$$y(t) = \begin{cases} t-3, & 1 \le t \le 5, \\ 0, & \text{otherwise}. \end{cases}$$

Since $x(t)$ is a unit rectangular pulse of width 2,

$$z(t) = x(t) * y(t) = \int_{-1}^{1} y(t-\tau)\, d\tau = \int_{t-1}^{t+1} y(u)\, du.$$

Thus, $z(t)$ equals the area under $y(u)$ over a sliding window of width 2.
Evaluating,

$$z(t) = \begin{cases} \frac{t^2}{2} - 2t + \frac{3}{2}, & 0 \le t \le 2, \\ 2t - 6, & 2 \le t \le 4, \\ -\frac{t^2}{2} + 4t - 4, & 4 \le t \le 6. \end{cases}$$

For the last interval,

$$\frac{dz(t)}{dt} = 4 - t.$$

Hence,

$$\frac{dz(t)}{dt} = 0 \quad \Rightarrow \quad T_1 = 4 \text{ s}.$$

4

Q2.22.1 MCQ | 2022 | 2M | Ans: B | ● ○ ○

The system is defined by

$$y(t) = x\left(e^{t}\right).$$

Linearity:
Let

$$x(t) = a\,x_1(t) + b\,x_2(t).$$

Then,

$$\begin{aligned} y(t) &= x(e^{t}) \\ &= a\,x_1(e^{t}) + b\,x_2(e^{t}) \\ &= a\,y_1(t) + b\,y_2(t). \end{aligned}$$

Hence, the system is **linear**.

Causality:
At time $t = t_0$,

$$y(t_0) = x(e^{t_0}).$$

Since

$$e^{t_0} > t_0, \qquad \forall\, t_0,$$

the output depends on the value of the input at a future time. Therefore, the system is **non-causal**.

(b) Linear and non-causal.

Q2.22.2 MCQ | 2022 | 2M | Ans: D | ● ○ ○

A causal LTI system is governed by

$$y(t) + \frac{1}{4}\frac{dy(t)}{dt} = 2x(t).$$

Taking the Laplace transform under zero initial conditions,

$$Y(s) + \frac{s}{4}Y(s) = 2X(s).$$

Hence, the transfer function is

$$H(s) = \frac{Y(s)}{X(s)} = \frac{2}{1 + \frac{s}{4}} = \frac{8}{s+4}.$$

Taking the inverse Laplace transform,

$$h(t) = 8e^{-4t}u(t).$$

(d) $8e^{-4t}u(t)$

Q2.21.1 MCQ | 2021 | 1M | Ans: C | ● ○ ○

The impulse response of cascaded discrete-time LTI systems is the convolution of their individual impulse responses:

$$h[n] = h_1[n] * h_2[n].$$

Given,

$$h_1[n] = \delta[n-1] + \delta[n+1],$$

and

$$h_2[n] = \delta[n] + \delta[n-1].$$

Using the distributive property,

$$\begin{aligned} h[n] &= (\delta[n-1] + \delta[n+1]) * (\delta[n] + \delta[n-1]) \\ &= \delta[n-1] * \delta[n] + \delta[n-1] * \delta[n-1] \\ &\quad + \delta[n+1] * \delta[n] + \delta[n+1] * \delta[n-1]. \end{aligned}$$

Using

$$\delta[n-a] * \delta[n-b] = \delta[n-(a+b)],$$

we obtain

$$\begin{aligned} \delta[n-1] * \delta[n] &= \delta[n-1], \\ \delta[n-1] * \delta[n-1] &= \delta[n-2], \\ \delta[n+1] * \delta[n] &= \delta[n+1], \\ \delta[n+1] * \delta[n-1] &= \delta[n]. \end{aligned}$$

Hence,

$$h[n] = \delta[n-2] + \delta[n-1] + \delta[n] + \delta[n+1].$$

$$\boxed{\text{(c) } \delta[n-2] + \delta[n-1] + \delta[n] + \delta[n+1]}$$

Q2.21.2 MCQ | 2021 | 1M | Ans: D | ● ○ ○

The system is defined by

$$y(t) = \max(0, x(t)).$$

Linearity:

Consider the input

$$x(t) = 1.$$

Then,

$$T\{x(t)\} = 1.$$

For the scaled input,

$$-x(t) = -1,$$

we get

$$T\{-x(t)\} = \max(0, -1) = 0.$$

However,

$$-T\{x(t)\} = -1 \neq 0.$$

Hence, the homogeneity property is violated. Therefore, the system is **non-linear**.

Time Invariance:

Let the input be delayed by t_0:

$$x_1(t) = x(t - t_0).$$

The corresponding output is

$$y_1(t) = \max(0, x(t - t_0)) .$$

Now delay the original output:

$$y(t - t_0) = \max(0, x(t - t_0)) .$$

Since

$$y_1(t) = y(t - t_0),$$

the system is **time-invariant**.

(d) Non-linear and time-invariant.

Q2.20.1 MCQ 2020 2M Ans: C ● ○ ○

The given LTI system satisfies the difference equation

$$y[n] - ay[n - 1] = bx[n] - bx[n - 1].$$

Taking the z-transform under zero initial conditions,

$$Y(z)\left(1 - az^{-1}\right) = X(z)\, b\left(1 - z^{-1}\right).$$

Hence,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b(1 - z^{-1})}{1 - az^{-1}}.$$

For $a \neq 0$,

$$\frac{1}{1 - az^{-1}} = \sum_{k=0}^{\infty} a^k z^{-k}.$$

Therefore,

$$h[n] = b\delta[n] + b(a - 1)a^{n-1}u[n - 1].$$

Since the exponential term is non-zero for infinitely many values of n, the impulse response is infinite in duration. Hence, Option C is true.
Check the remaining options.

- **A:** False. Due to the recursive term $y[n - 1]$, the output depends on all past inputs through the recursion.
- **B:** False. A difference equation alone does not guarantee causality; different regions of convergence may correspond to causal or non-causal systems.
- **D:** False. Even if $x[n] = 0$ for $n < 0$, the output for $n > 0$ also depends on the initial condition $y[-1]$. Hence it is not solely determined by $x[n]$.

(c) The system impulse response is non-zero at infinitely many instants.

Q2.20.2 MCQ 2020 2M Ans: C ● ○ ○

For the given LTI system,

$$y(t) = x(t) * h(t),$$

and

$$z(t) = |x(t)| * |h(t)|.$$

Using the definition of convolution,

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau.$$

Taking the magnitude,

$$|y(t)| = \left| \int_{-\infty}^{\infty} x(\tau) h(t-\tau)\, d\tau \right|.$$

Applying the triangle inequality,

$$|y(t)| \le \int_{-\infty}^{\infty} |x(\tau)|\, |h(t-\tau)|\, d\tau = z(t).$$

Thus,

$$z(t) \ge |y(t)| \ge y(t),$$

for every t.
Hence,

$$z(t) \ge y(t), \qquad \forall\, t.$$

(c) For all $t \in (-\infty, \infty),\ z(t) \ge y(t).$

Key Points

- Triangle inequality:

$$\left| \int f(\tau)\, d\tau \right| \le \int |f(\tau)|\, d\tau.$$

- For any LTI system,

$$|x * h| \le |x| * |h|.$$

Q2.20.3 MCQ | 2020 | 1M | Ans: A | ● ○ ○

The given LTI system is described by

$$\frac{d^2 y(t)}{dt^2} + 4y(t) = 6r(t).$$

Taking the Laplace transform under zero initial conditions,

$$s^2 Y(s) + 4Y(s) = 6R(s).$$

Hence, the transfer function is

$$H(s) = \frac{Y(s)}{R(s)} = \frac{6}{s^2 + 4}.$$

The poles are obtained by setting the denominator equal to zero:

$$s^2 + 4 = 0.$$

Therefore,

$$s = \pm 2j.$$

(a) $+2j, -2j$

Q2.19.1 MCQ 2019 1M Ans: D ● ○ ○

For a causal LTI system, the impulse response must satisfy

$$h(t) = 0, \qquad t < 0.$$

Examine each option.

- **A:**

$$h(t) = e^{-a(t-T)}u(t).$$

  Since $u(t) = 0$ for $t < 0$, the impulse response is causal.

- **B:**

$$h(t) = e^{-a(t+T)}u(t) = e^{-aT}e^{-at}u(t).$$

  Again, $u(t)$ ensures

$$h(t) = 0, \quad t < 0,$$

  so it is causal.

- **C:**

$$h(t) = e^{at}u(t).$$

  Although the response is unstable for $a > 0$, it is zero for $t < 0$. Hence, it is still a valid causal impulse response.

- **D:**

$$h(t) = 1 + e^{-at}u(t).$$

  For $t < 0$,

$$h(t) = 1 \neq 0,$$

  which violates the causality condition.

Hence, Option D cannot be the impulse response of a causal LTI system.

(d) $1 + e^{-at}u(t)$

Q2.18.1 MCQ 2018 1M Ans: A ● ○ ○

An input $x(t)$ is called an eigenfunction of an LTI system if the output is a scaled version of the same input.
Let

$$y(t) = T\{x(t)\}.$$

If $x(t)$ is an eigenfunction of the LTI system, then

$$y(t) = k\,x(t),$$

where k is a constant called the *eigenvalue*.
Hence, the output differs only by a constant scaling factor, while the input waveform remains unchanged.

(a) $k\,x(t)$, where k is an eigenvalue.

Q2.17.1 MCQ 2017 1M Ans: D ● ○ ○

The given discrete-time system is

$$y[n] = (1 + (-1)^n)\, x[n].$$

Invertibility:
Since

$$1 + (-1)^n = \begin{cases} 2, & n \text{ even}, \\ 0, & n \text{ odd}, \end{cases}$$

the output becomes

$$y[n] = \begin{cases} 2x[n], & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}$$

Thus, all odd-indexed samples of $x[n]$ are lost in the output and cannot be recovered. Hence, the system is **non-invertible**.

Time Invariance:
Let the input be delayed by n_0:

$$x_1[n] = x[n - n_0].$$

The corresponding output is

$$y_1[n] = (1 + (-1)^n)\, x[n - n_0].$$

Now delay the original output:

$$y[n - n_0] = (1 + (-1)^{n-n_0})\, x[n - n_0].$$

Since, in general,

$$(-1)^n \neq (-1)^{n-n_0},$$

we have

$$y_1[n] \neq y[n - n_0].$$

Therefore, the system is **time-varying**.

(d) Non-invertible and time-varying.

Q2.17.2 NAT 2017 2M Ans: 0.098 ● ○ ○

The poles of the causal and stable system are

$$p_k = \frac{1}{\sqrt{2}} \exp\left(j \frac{(2k-1)\pi}{4}\right), \qquad k = 1, 2, 3, 4,$$

and all the zeros are at $z = 0$.
Since all four zeros are at the origin,

$$H(z) = K \frac{z^4}{\prod_{k=1}^{4} (z - p_k)}.$$

The poles are

$$\frac{1+j}{2}, \frac{-1+j}{2}, \frac{-1-j}{2}, \frac{1-j}{2}.$$

Hence,

$$\prod_{k=1}^{4} (z - p_k) = \left(z^2 - z + \frac{1}{2}\right)\left(z^2 + z + \frac{1}{2}\right) = z^4 + \frac{1}{4}.$$

Therefore,

$$H(z) = K\frac{z^4}{z^4 + \frac{1}{4}} = \frac{K}{1 + \frac{1}{4}z^{-4}}.$$

Using

$$H(1) = \frac{5}{4},$$

we get

$$\frac{K}{1 + \frac{1}{4}} = \frac{5}{4} \quad \Rightarrow \quad K = \frac{25}{16}.$$

Thus,

$$H(z) = \frac{\frac{25}{16}}{1 + \frac{1}{4}z^{-4}} = \frac{25}{16}\sum_{m=0}^{\infty}\left(-\frac{1}{4}\right)^m z^{-4m}.$$

Hence,

$$h[n] = \frac{25}{16}\sum_{m=0}^{\infty}\left(-\frac{1}{4}\right)^m \delta[n - 4m].$$

Since

$$g[n] = j^n h[n],$$

at $n = 8$,

$$h[8] = \frac{25}{16}\left(-\frac{1}{4}\right)^2 = \frac{25}{256},$$

and

$$j^8 = 1.$$

Therefore,

$$g[8] = \frac{25}{256} = 0.09765625.$$

$$\boxed{0.098}$$

Q2.17.3 MCQ 2017 2M Ans: B

For an LTI system with zero initial conditions,

$$\frac{d^2y(t)}{dt^2} + 7\frac{dy(t)}{dt} + 10y(t) = 4x(t) + 5\frac{dx(t)}{dt}$$

Taking Laplace transform,

$$(s^2 + 7s + 10)Y(s) = (4 + 5s)X(s)$$

Hence, the transfer function is

$$H(s) = \frac{Y(s)}{X(s)} = \frac{5s + 4}{(s + 2)(s + 5)}.$$

Using partial fraction expansion,

$$\frac{5s + 4}{(s + 2)(s + 5)} = \frac{A}{s + 2} + \frac{B}{s + 5}$$

$$5s + 4 = A(s + 5) + B(s + 2).$$

Putting $s = -2$,

$$-6 = 3A \Rightarrow A = -2.$$

Putting $s = -5$,

$$-21 = -3B \Rightarrow B = 7.$$

Therefore,

$$H(s) = \frac{-2}{s+2} + \frac{7}{s+5}.$$

Taking inverse Laplace transform,

$$h(t) = \left(-2e^{-2t} + 7e^{-5t}\right) u(t).$$

$$\boxed{\text{(b) } -2e^{-2t}u(t) + 7e^{-5t}u(t)}$$

Q2.17.4 MCQ 2017 2M Ans: D ● ○ ○

Since the overall impulse response of a cascade system is the convolution of the individual impulse responses,

$$h(n) = h_1(n) * h_2(n)$$

where

$$h_1(n) = \{1, -1\}, \qquad h_2(n) = \{1, 1\}.$$

Convolving the two sequences,

$$h(n) = \{1, -1\} * \{1, 1\} = \{1, 0, -1\}.$$

Hence,

$$y(n) = x(n) * h(n) = x(n) * \{1, 0, -1\}.$$

This is equivalent to

$$y(n) = x(n) - x(n-2).$$

Now test the options.
For Option D,

$$x(n) = \{1, 2, 2, 1\},$$

$$y(n) = \{1, 2, 1, -1, -2, -1\},$$

which exactly matches the given output sequence.

$$\boxed{\text{(d) } x(n) = \{1, 2, 2, 1\}}$$

Q2.16.1 MCQ 2016 1M Ans: C ● ○ ○

The given system is

$$y(t) = x(t)\cos(t).$$

Linearity:
Let the input be

$$x(t) = a\,x_1(t) + b\,x_2(t).$$

Then,

$$\begin{aligned} y(t) &= [a\,x_1(t) + b\,x_2(t)]\cos(t) \\ &= a\,x_1(t)\cos(t) + b\,x_2(t)\cos(t) \\ &= a\,y_1(t) + b\,y_2(t). \end{aligned}$$

Hence, the system satisfies the superposition principle and is **linear**.
Time Invariance:
For input $x(t - t_0)$,

$$y_1(t) = x(t - t_0)\cos(t).$$

Delaying the original output by t_0,

$$y(t - t_0) = x(t - t_0)\cos(t - t_0).$$

Since

$$x(t - t_0)\cos(t) \neq x(t - t_0)\cos(t - t_0),$$

the delayed input does not produce a delayed output.
Therefore, the system is **time-varying**.

(c) Linear and time-varying

Key Points

- Multiplication by a known function of time preserves linearity.
- If the system equation explicitly contains t, it is generally time-varying.

Mistakes to be avoided

- Do not confuse multiplication by $\cos(t)$ with a nonlinear operation.
- Presence of an explicit time-dependent coefficient indicates a time-varying system.

Q2.15.1 MCQ 2015 2M Ans: D ● ○ ○

For a continuous-time LTI system, the necessary and sufficient condition for BIBO stability is

$$\int_{-\infty}^{\infty} |h(t)|\,dt < \infty,$$

i.e., the impulse response must be **absolutely integrable**.
Since the input $u(t)$ is bounded ($|u(t)| \leq 1$), the corresponding output is the step response

$$s(t) = h(t) * u(t).$$

For a BIBO stable system,

$$|s(t)| = \left| \int_{-\infty}^{t} h(\tau)\, d\tau \right| \leq \int_{-\infty}^{\infty} |h(\tau)|\, d\tau < \infty.$$

Hence, the unit step response is always **bounded**.
Checking the options:

A. False: BIBO stability requires *absolute* integrability of $h(t)$.

B. False: The impulse response need not have finite support.

C. False: The step response is not necessarily absolutely integrable.

D. True: The unit step response is bounded.

(d) The unit step response will be bounded.

Q2.15.2 MCQ 2015 2M Ans: D ● ○ ○

The given impulse response is

$$g(t) = \begin{cases} 1, & 0 \leq t \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

For two cascaded LTI systems, the overall impulse response is the convolution

$$h(t) = g(t) * g(t).$$

Evaluating the convolution,

$$h(t) = \begin{cases} t, & 0 \leq t \leq 1, \\ 2 - t, & 1 \leq t \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

This is a triangular waveform. Its maximum value occurs at

$$t = 1,$$

giving

$$h_{\max} = 1.$$

(d) 1

Q2.14.1 MCQ 2014 1M Ans: C ● ○ ○

Let

$$x(t) \neq 0 \quad \text{only for} \quad T_x < t < T_x',$$

and

$$y(t) \neq 0 \quad \text{only for} \quad T_y < t < T_y'.$$

The convolution is

$$z(t) = x(t) * y(t).$$

The support of the convolution is obtained by adding the support intervals of the two signals.
Hence,

$$T_x + T_y < t < T_x' + T_y'.$$

Therefore,

$$z(t) = 0$$

outside this interval.
Thus, only Option C is correct.

(c) $z(t)$ is zero outside $T_x + T_y < t < T_x' + T_y'$.

Q2.13.1 MCQ 2013 2M Ans: B ● ○ ○

The step response of an LTI system is the response to the unit step input:

$$s(t) = h(t) * u(t).$$

Given,

$$h(t) = \delta(t-1) + \delta(t-3).$$

Using the property

$$\delta(t-a) * u(t) = u(t-a),$$

the step response is

$$s(t) = u(t-1) + u(t-3).$$

At $t = 2$,

$$s(2) = u(1) + u(-1) = 1 + 0 = 1.$$

(b) 1

Q2.13.2 MCQ 2013 1M Ans: C ● ○ ○

For two LTI systems connected in cascade, the overall impulse response is obtained by convolving the individual impulse responses.
Let the impulse responses be $h_1(t)$ and $h_2(t)$.
The overall impulse response is

$$h(t) = h_1(t) * h_2(t) = \int_{-\infty}^{\infty} h_1(\tau)\, h_2(t-\tau)\, d\tau.$$

Hence, the cascade connection is represented by the convolution of the two impulse responses.

(c) Convolution of $h_1(t)$ and $h_2(t)$

Q2.12.1 MCQ | 2012 | 2M | Ans: D

The input-output relationship is

$$y(t) = \int_{-\infty}^{t} x(\tau)\cos(3\tau)\,d\tau.$$

Time Invariance:

For a delayed input $x(t - t_0)$,

$$y_1(t) = \int_{-\infty}^{t} x(\tau - t_0)\cos(3\tau)\,d\tau.$$

Let $u = \tau - t_0$. Then,

$$y_1(t) = \int_{-\infty}^{t-t_0} x(u)\cos(3u + 3t_0)\,du.$$

Since

$$\cos(3u + 3t_0) \neq \cos(3u),$$

we have

$$y_1(t) \neq y(t - t_0).$$

Hence, the system is **not time-invariant**.

Stability:

Consider the bounded input

$$x(t) = \cos(3t)\,u(t),$$

where $u(t)$ is the unit step function. Since

$$|x(t)| = |\cos(3t)u(t)| \leq 1,$$

the input is bounded.

The output is

$$y(t) = \int_{-\infty}^{t} x(\tau)\cos(3\tau)\,d\tau.$$

Substituting $x(\tau) = \cos(3\tau)u(\tau)$,

$$y(t) = \begin{cases} 0, & t < 0, \\ \displaystyle\int_0^t \cos^2(3\tau)\,d\tau, & t \geq 0. \end{cases}$$

Using the identity

$$\cos^2\theta = \frac{1 + \cos(2\theta)}{2},$$

we obtain

$$\begin{aligned} y(t) &= \int_0^t \frac{1 + \cos(6\tau)}{2}\,d\tau \\ &= \frac{1}{2}\int_0^t d\tau + \frac{1}{2}\int_0^t \cos(6\tau)\,d\tau \\ &= \frac{t}{2} + \frac{\sin(6t)}{12}, \qquad t \geq 0. \end{aligned}$$

Hence,

$$y(t) = \begin{cases} 0, & t < 0, \\ \dfrac{t}{2} + \dfrac{\sin(6t)}{12}, & t \geq 0. \end{cases}$$

Since the output contains the ramp term

$$\frac{t}{2},$$

it grows unbounded as $t \to \infty$, i.e.,

$$|y(t)| \to \infty.$$

Thus, a bounded input produces an unbounded output. Therefore, the system is **not BIBO stable**.

(d) Not time-invariant and not stable

Q2.12.2 MCQ 2012 2M Ans: A ●○○

Method 1

The output is the convolution of $h[n]$ and $g[n]$:

$$y[n] = h[n] * g[n] = \sum_{k=-\infty}^{\infty} h[k]\, g[n-k].$$

Given,

$$h[n] = \left(\frac{1}{2}\right)^n u[n],$$

and $g[n]$ is causal, i.e.,

$$g[n] = 0, \qquad n < 0.$$

For $n = 0$,

$$y[0] = h[0]g[0] = 1 \cdot g[0].$$

Since $y[0] = 1$,

$$g[0] = 1.$$

Now for $n = 1$,

$$y[1] = h[0]g[1] + h[1]g[0]$$
$$= g[1] + \frac{1}{2}(1).$$

Given,

$$y[1] = \frac{1}{2},$$

therefore,

$$g[1] + \frac{1}{2} = \frac{1}{2}$$

$$\Rightarrow \quad g[1] = 0.$$

Method 2

The convolution is

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\, g[n-k].$$

Since

$$h[n] = \left(\frac{1}{2}\right)^n u[n],$$

its initial samples are

$$h[0] = 1, \qquad h[1] = \frac{1}{2}, \qquad h[2] = \frac{1}{4}, \ldots$$

Also, $g[n]$ is causal, so

$$g[n] = 0, \qquad n < 0.$$

Using the convolution matrix,

$$\begin{bmatrix} y[0] \\ y[1] \end{bmatrix} = \begin{bmatrix} h[0] & 0 \\ h[1] & h[0] \end{bmatrix} \begin{bmatrix} g[0] \\ g[1] \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} g[0] \\ g[1] \end{bmatrix}.$$

Since

$$y[0] = 1, \qquad y[1] = \frac{1}{2},$$

we obtain

$$\begin{bmatrix} 1 \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} g[0] \\ g[1] \end{bmatrix}.$$

From the first equation,

$$g[0] = 1.$$

Substituting into the second equation,

$$\frac{1}{2} = \frac{1}{2}(1) + g[1].$$

Hence,

$$g[1] = 0.$$

(a) 0

Q2.11.1 MCQ 2011 2M Ans: C ● ○ ○

The step response of an LTI system is obtained by convolving the impulse response with the unit step function:

$$s(t) = h(t) * u(t).$$

Given,

$$h(t) = \left(e^{-t} + e^{-2t}\right) u(t).$$

The step response is

$$s(t) = \int_0^t \left(e^{-\tau} + e^{-2\tau}\right) d\tau.$$

Evaluating the integral,

$$s(t) = \left(1 - e^{-t}\right) + \frac{1}{2}\left(1 - e^{-2t}\right)$$
$$= \frac{3}{2} - e^{-t} - \frac{1}{2}e^{-2t}.$$

Hence,

$$s(t) = \left(\frac{3}{2} - e^{-t} - \frac{1}{2}e^{-2t}\right)u(t).$$

(c) $\left(1.5 - e^{-t} - 0.5e^{-2t}\right)u(t)$

Q2.10.1 MCQ 2010 2M Ans: C ●○○

Given,

$$x[n] = \{1, -1\},$$

and

$$y[n] = \{1, 0, 0, 0, -1\}.$$

Since

$$y[n] = x[n] * h[n],$$

we determine the impulse response $h[n]$.
Using the convolution with

$$x[n] = \delta[n] - \delta[n-1],$$

we get

$$y[n] = h[n] - h[n-1].$$

Now compute the samples sequentially:

$$y[0] = h[0] = 1$$

$$y[1] = h[1] - h[0] = 0 \Rightarrow h[1] = 1$$

$$y[2] = h[2] - h[1] = 0 \Rightarrow h[2] = 1$$

$$y[3] = h[3] - h[2] = 0 \Rightarrow h[3] = 1.$$

Finally,

$$y[4] = -h[3] = -1,$$

which is satisfied.
Hence,

$$h[n] = \{1, 1, 1, 1\}.$$

(c) $h[n] = \{1, 1, 1, 1\}$

Q2.10.2 MCQ 2010 1M Ans: B ● ○ ○

The system is described by

$$y(t) = \int_{-\infty}^{5t} x(\tau)\, d\tau, \qquad t > 0.$$

Linearity:
Let

$$x(t) = a\, x_1(t) + b\, x_2(t).$$

Then,

$$\begin{aligned} y(t) &= \int_{-\infty}^{5t} [a\, x_1(\tau) + b\, x_2(\tau)]\, d\tau \\ &= a \int_{-\infty}^{5t} x_1(\tau) d\tau + b \int_{-\infty}^{5t} x_2(\tau) d\tau \\ &= a\, y_1(t) + b\, y_2(t). \end{aligned}$$

Hence, the system is **linear**.
Causality:
The output at time t depends on

$$x(\tau), \qquad -\infty < \tau \le 5t.$$

Since $5t > t$ for $t > 0$, the output depends on future values of the input ($t < \tau \le 5t$).
Therefore, the system is **non-causal**.

(b) Linear but not causal

Q2.09.1 MCQ 2009 2M Ans: B ● ○ ○

A cascade of three LTI systems is given to be **causal** and **unstable**.
Causality:
A cascade is causal only if each individual system is causal. If even one system is non-causal, the overall cascade becomes non-causal.
Hence, all three systems must be causal.
Stability:
For a cascade of LTI systems to be BIBO stable, every individual system must be stable. Since the overall cascade is unstable, at least one of the systems must be unstable.
Therefore,

At least one system is unstable and at least one system is causal.

Checking the options:

A. False: Every system need not be unstable.

B. True: At least one system is unstable, and since all systems are causal, at least one system is certainly causal.

C. False: It is not necessary that all systems are unstable.

D. False: No conclusion can be made about the majority being unstable.

(b) At least one system is unstable and at least one system is causal

Q2.09.2 MCQ 2009 1M Ans: D ● ○ ○

For the original LTI system,

$$y(t) = x(t) * h(t).$$

Now the modified input and impulse response are

$$x_1(t) = x(t-\tau), \qquad h_1(t) = h(t-\tau).$$

The new output is

$$\begin{aligned} y_1(t) &= x_1(t) * h_1(t) \\ &= \int_{-\infty}^{\infty} x(\lambda-\tau)\, h(t-\tau-\lambda)\, d\lambda. \end{aligned}$$

Let

$$u = \lambda - \tau \quad \Rightarrow \quad d\lambda = du.$$

Then,

$$\begin{aligned} y_1(t) &= \int_{-\infty}^{\infty} x(u)\, h(t-2\tau-u)\, du \\ &= y(t-2\tau). \end{aligned}$$

Thus, delaying both the input and the impulse response by τ delays the output by 2τ.

(d) $y(t-2\tau)$

Q2.08.1 MCQ 2008 2M Ans: D ● ○ ○

The system is defined by

$$y(t) = \int_{-\infty}^{-2t} x(\tau)\, d\tau.$$

We examine its causality, time invariance and stability.

Causality:

The output at time t depends on the input over

$$-\infty < \tau \le -2t.$$

For $t < 0$,

$$-2t > t,$$

so the output depends on future values of the input. Hence, the system is **non-causal**.

Time Invariance:

For the delayed input $x(t-t_0)$,

$$y_1(t) = \int_{-\infty}^{-2t} x(\tau - t_0)\, d\tau.$$

Let $u = \tau - t_0$. Then,

$$y_1(t) = \int_{-\infty}^{-2t-t_0} x(u)\, du.$$

However,

$$y(t - t_0) = \int_{-\infty}^{-2(t-t_0)} x(u)\, du = \int_{-\infty}^{-2t+2t_0} x(u)\, du.$$

Since

$$-2t - t_0 \neq -2t + 2t_0,$$

we have

$$y_1(t) \neq y(t - t_0).$$

Therefore, the system is **time-varying**.

Stability:

For a bounded input $|x(t)| \leq M$,

$$|y(t)| \leq M \int_{-\infty}^{-2t} d\tau = \infty.$$

Thus, a bounded input can produce an unbounded output, so the system is **unstable**.

(d) Non-causal, time-varying and unstable

Q2.08.2 MCQ 2008 1M Ans: D ●○○

A causal linear time-varying (LTV) system has impulse response $h(t)$.

We examine the given statements.

Statement (I):

The principle of superposition is the defining property of a linear system. Since the system is stated to be *linear*, superposition holds.

Hence, Statement (I) is **correct**.

Statement (II):

For a causal system, the response cannot occur before the impulse is applied. Therefore, the impulse response must satisfy

$$h(t) = 0, \qquad t < 0.$$

Hence, Statement (II) is also **correct**.

(d) Both Statement (I) and Statement (II) are correct

Q2.08.3 MCQ 2008 1M Ans: A ● ○ ○

The input signal is

$$x(t) = e^{-\alpha t}\sin(\omega t).$$

Express the input as the imaginary part of a complex exponential:

$$x(t) = \Im\left\{e^{(-\alpha+j\omega)t}\right\}.$$

For an LTI system, the complex exponential

$$e^{st}$$

is an eigenfunction. Hence,

$$e^{st} \longrightarrow H(s)e^{st},$$

where $H(s)$ is the system transfer function evaluated at $s = -\alpha + j\omega$.
Therefore,

$$y(t) = |H(s)|\, e^{-\alpha t}\sin(\omega t + \angle H(s)).$$

Comparing with

$$Ke^{-\beta t}\sin(\nu t + \phi),$$

we obtain

$$\beta = \alpha, \qquad \nu = \omega.$$

(c) $\beta = \alpha$ and $\nu = \omega$

Q2.07.1 MCQ 2007 1M Ans: D ● ○ ○

A stable LTI system is excited by the sinusoidal input

$$x(t) = a_1\sin(\omega_1 t + \phi_1).$$

For a stable LTI system, a sinusoidal input produces a sinusoidal steady-state output of the *same frequency*. Only the amplitude and phase are modified by the system.
Hence,

$$y_{ss}(t) = a_2\sin(\omega_1 t + \phi_2),$$

or equivalently,

$$y_{ss}(t) = a_2\cos(\omega_1 t + \phi),$$

since sine and cosine differ only by a phase shift.
Therefore,

$$\omega_2 = \omega_1,$$

and $F(\cdot)$ must be either a sine or a cosine function.
Checking the options:

A. False: F cannot be an arbitrary periodic function.

B. False: The amplitude generally changes, so $a_1 \neq a_2$ in general.

C. False: The phase need not remain unchanged.

D. True: F must be a sine or cosine function with $\omega_1 = \omega_2$.

(d) F must be a "sine" or "cosine" function with $\omega_1 = \omega_2$.

Q2.06.1 MCQ | 2006 | 2M | Ans: A

Method 1

Given,

$$x[-1] = -1, \qquad x[0] = 2,$$

and all other samples of $x[n]$ are zero.
Also,

$$y[-1] = -1, \quad y[0] = 3, \quad y[1] = -1, \quad y[2] = -2,$$

with $y[n] = 0$ for $n < -1$ and $n > 2$.
Using the convolution relation,

$$y[n] = x[n] * h[n] = \sum_k x[k]h[n-k].$$

Since only $x[-1]$ and $x[0]$ are non-zero,

$$y[n] = -h[n+1] + 2h[n].$$

Now compute the impulse response recursively.
From $n = 2$,

$$2h[2] - h[3] = -2.$$

As $y[n] = 0$ for $n > 2$, we get $h[3] = 0$, hence

$$h[2] = -1.$$

Next,

$$2h[1] - h[2] = -1 \Rightarrow 2h[1] + 1 = -1 \Rightarrow h[1] = -1.$$

Similarly,

$$2h[0] - h[1] = 3 \Rightarrow 2h[0] + 1 = 3 \Rightarrow h[0] = 1.$$

Finally,

$$2h[-1] - h[0] = -1 \Rightarrow 2h[-1] - 1 = -1 \Rightarrow h[-1] = 0.$$

Also, using $y[n] = 0$ outside the given range gives

$$h[n] = 0, \qquad n < 0,\ n > 2.$$

Hence,

$$h[0] = 1, \qquad h[1] = h[2] = -1.$$

Method 2

Given

$$x[-1] = -1, \qquad x[0] = 2,$$

and all other samples of $x[n]$ are zero.
The convolution is

$$y[n] = x[n] * h[n].$$

Since only $x[-1]$ and $x[0]$ are non-zero,

$$y[n] = -h[n+1] + 2h[n].$$

Using the given output samples,

$$y[-1] = -1, \qquad y[0] = 3, \qquad y[1] = -1, \qquad y[2] = -2.$$

These equations can be written in matrix form as

$$\begin{bmatrix} y[-1] \\ y[0] \\ y[1] \\ y[2] \end{bmatrix} = \begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} h[-1] \\ h[0] \\ h[1] \\ h[2] \end{bmatrix}.$$

Substituting the given output values,

$$\begin{bmatrix} -1 \\ 3 \\ -1 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} h[-1] \\ h[0] \\ h[1] \\ h[2] \end{bmatrix}.$$

Since the coefficient matrix is upper triangular, solve from the last equation upward:

$$2h[2] = -2 \quad \Rightarrow \quad h[2] = -1,$$

$$2h[1] - h[2] = -1 \quad \Rightarrow \quad 2h[1] + 1 = -1 \quad \Rightarrow \quad h[1] = -1,$$

$$2h[0] - h[1] = 3 \quad \Rightarrow \quad 2h[0] + 1 = 3 \quad \Rightarrow \quad h[0] = 1,$$

$$2h[-1] - h[0] = -1 \quad \Rightarrow \quad 2h[-1] - 1 = -1 \quad \Rightarrow \quad h[-1] = 0.$$

Also, since

$$y[n] = 0, \qquad n < -1 \text{ and } n > 2,$$

it follows that

$$h[n] = 0, \qquad n < 0,\ n > 2.$$

$$\boxed{\text{(a) } h[n] = 0 \text{ for } n < 0,\ n > 2,\ h[0] = 1,\ h[1] = h[2] = -1}$$

Q2.06.2 MCQ 2006 2M Ans: D ● ○ ○

The system is defined as

$$y(t) = e^{-|x(t)|}.$$

We determine the condition under which the output remains bounded.
Since

$$|x(t)| \geq 0,$$

we have

$$0 < e^{-|x(t)|} \leq 1$$

for every finite or infinite value of $x(t)$.
Thus,

$$0 < y(t) \leq 1,$$

which shows that the output is always bounded irrespective of whether $x(t)$ is bounded or unbounded.
Hence, $y(t)$ remains bounded even when $x(t)$ is unbounded.
Checking the options:

A. False: Bounded input is not necessary.

B. False: $x(t)$ need not be non-negative because $|x(t)|$ is used.

C. False: The output is bounded for all t, not only for $t \geq 0$.

D. True: The output is bounded even when $x(t)$ is not bounded.

(d) $y(t)$ is bounded even when $x(t)$ is not bounded.

Q2.02.1 MCQ 2002 1M Ans: B ● ○ ○

Let $s(t)$ denote the step response of a linear system with zero initial conditions.
Then,

$$s(t) = h(t) * u(t),$$

where $h(t)$ is the impulse response.
The response of the system to the input $u(t)$ is

$$y(t) = h(t) * u(t).$$

Since

$$s(t) = h(t) * u(t),$$

we have

$$y(t) = s(t).$$

Now,

$$\frac{d}{dt}s(t) = \frac{d}{dt}\left[\int_0^t s(t-\tau)u(\tau)\,d\tau\right],$$

because convolution with the unit impulse is obtained by differentiating the convolution with the unit step:

$$\frac{d}{dt}(s * u) = s * \delta = s.$$

Hence, the required response is represented by

$$\boxed{\frac{d}{dt}\left[\int_0^t s(t-\tau)u(\tau)\,d\tau\right]}.$$

$$\boxed{\text{(b) } \frac{d}{dt}\left[\int_0^t s(t-\tau)u(\tau)\,d\tau\right]}$$

Q2.93.1 MCQ | 1993 | 1M | Ans: A | ● ○ ○

Let $h(t)$ be the impulse response and $s(t)$ be the step response of the system.
Since

$$s(t) = h(t) * u(t),$$

the impulse response is

$$h(t) = \frac{ds(t)}{dt}.$$

For any input $u(t)$, the output is

$$y(t) = h(t) * u(t).$$

Substituting $h(t) = \frac{ds(t)}{dt}$,

$$y(t) = \frac{ds(t)}{dt} * u(t).$$

Using the differentiation property of convolution,

$$\frac{ds(t)}{dt} * u(t) = \frac{d}{dt}\left[s(t) * u(t)\right].$$

Since

$$s(t) * u(t) = \int_0^t s(t-\tau)u(\tau)\,d\tau,$$

we obtain

$$y(t) = \frac{d}{dt}\left[\int_0^t s(t-\tau)u(\tau)\,d\tau\right].$$

$$\boxed{\text{(a) } \frac{d}{dt}\left[\int_0^t s(t-\tau)u(\tau)\,d\tau\right]}$$

Q2.92.1 NAT | 1992 | 5M | Ans: 2.8 | ● ○ ○

Given,

Method 1

$$h(t) = \begin{cases} 1, & 0 \le t < 1, \\ 0, & \text{otherwise.} \end{cases}$$

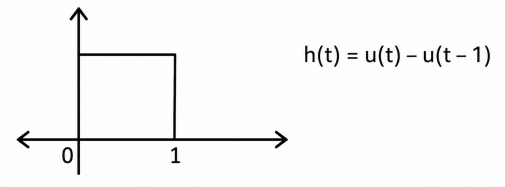


Fig. Solution Q2.92.1

For two identical networks connected in cascade, the overall impulse response is

$$h_{eq}(t) = h(t) * h(t).$$

$$h_{eq}(t) = \int_{-\infty}^{\infty} h(\tau)\, h(t-\tau)\, d\tau.$$

The value of the convolution equals the overlap length of the two rectangular pulses.
Hence,

$$h_{eq}(t) = \begin{cases} 0, & t < 0, \\ t, & 0 \le t < 1, \\ 2 - t, & 1 \le t < 2, \\ 0, & t \ge 2. \end{cases}$$

Method 2

Taking the Laplace transform,

$$H(s) = \frac{1 - e^{-s}}{s}$$

Hence,

$$G(s) = H(s)\, H(s) = \left(\frac{1 - e^{-s}}{s}\right)^2 = \frac{1}{s^2} + \frac{e^{-2s}}{s^2} - \frac{2e^{-s}}{s^2}$$

Taking the inverse Laplace transform,

$$g(t) = t\, u(t) + (t-2)u(t-2) - 2(t-1)u(t-1)$$

Thus,

$$g(t) = \begin{cases} 0, & t < 0, \\ t, & 0 \le t < 1, \\ 2 - t, & 1 \le t < 2, \\ 0, & t \ge 2. \end{cases}$$

This is a triangular waveform of peak value 1 at $t = 1$.

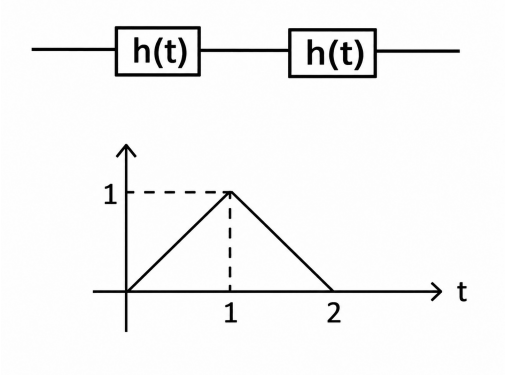


Fig. Solution Q2.92.1

Key Points

- The impulse response of cascaded LTI systems is the convolution of the individual impulse responses.
- Convolution of two unit-height rectangular pulses of width 1 results in a triangular pulse.
- The support of the resulting impulse response is $0 \leq t \leq 2$.

PrepFusion

SOLUTIONS

III

Continuous Time Fourier Series

Topics

1. Introduction to Fourier Series
2. Exponential Fourier Series
3. Predicting the Nature of a Signal from Fourier Coefficients
4. Power Using Fourier Series Coefficients
5. Properties of Exponential Fourier Series
6. Fourier Series of Periodic Waveforms
7. Trigonometric Fourier Series and Symmetry
8. Polar Form, Dirichlet's Conditions and Gibbs Phenomenon

Unit Snapshot*

Stream: EE	**Questions:** 21	**MCQ:** 20	**MSQ:** 0	**NAT:** 1
1M: 9	**2M:** 10	**Descriptive:** 2	**Avg Weightage** ~2M	**Range:** 1M(2017)–2M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 MCQ 2026 2M Ans: B ● ○ ○

The waveform is

$$g(x) = \begin{cases} -k, & -\pi < x \le 0, \\ +k, & 0 < x \le \pi, \\ 0, & \text{otherwise.} \end{cases}$$

The periodic signal is

$$f(x) = \sum_{m=-\infty}^{\infty} g(x + 2\pi m),$$

which is an odd square wave. Hence,

$$a_0 = 0, \qquad a_n = 0.$$

For the sine coefficients,

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx = \frac{2}{\pi}\int_{0}^{\pi} k\sin(nx)\,dx.$$

Therefore,

$$b_n = \frac{2k}{\pi}\left[\frac{1-\cos(n\pi)}{n}\right].$$

For odd n,

$$b_n = \frac{4k}{n\pi}.$$

Hence, for the third harmonic,

$$a_3 + b_3 = 0 + \frac{4k}{3\pi}.$$

Given,

$$\frac{4k}{3\pi} = \frac{2}{3\pi},$$

which gives

$$k = \frac{1}{2}.$$

$$\boxed{\text{(b) } \frac{1}{2}}$$

Key Points

- An odd periodic signal has only sine terms in its trigonometric Fourier series.
- For a square wave of amplitude k, $b_n = \frac{4k}{n\pi}$ for odd n and $b_n = 0$ for even n.

Mistakes to be avoided

- Do not compute cosine coefficients for an odd function.
- Remember that only odd harmonics are present in this square wave.

Q3.19.1 MCQ 2019 2M Ans: A ● ○ ○

Given,

$$f(t) = \begin{cases} A\sin t, & 0 \le t \le \pi, \\ 0, & \pi < t < 2\pi. \end{cases}$$

The Fourier series is

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos nt + \sum_{n=1}^{\infty} b_n \sin nt.$$

The first cosine coefficient is

$$a_1 = \frac{1}{\pi}\int_0^{2\pi} f(t)\cos t\, dt = \frac{A}{\pi}\int_0^{\pi} \sin t \cos t\, dt.$$

Using

$$\sin t \cos t = \frac{1}{2}\sin 2t,$$

we get

$$a_1 = \frac{A}{2\pi}\int_0^{\pi} \sin 2t\, dt = 0.$$

The first sine coefficient is

$$b_1 = \frac{1}{\pi}\int_0^{2\pi} f(t)\sin t\, dt = \frac{A}{\pi}\int_0^{\pi} \sin^2 t\, dt.$$

Since

$$\int_0^{\pi} \sin^2 t\, dt = \frac{\pi}{2},$$

we obtain

$$b_1 = \frac{A}{2}.$$

$$\boxed{\text{(a) } a_1 = 0,\ b_1 = \frac{A}{2}}$$

Q3.17.1 NAT 2017 1M Ans: MTA ●○○

The function has positive sawtooth segments for $t > 0$ and negative sawtooth segments for $t < 0$, apart from its zeros at integers.

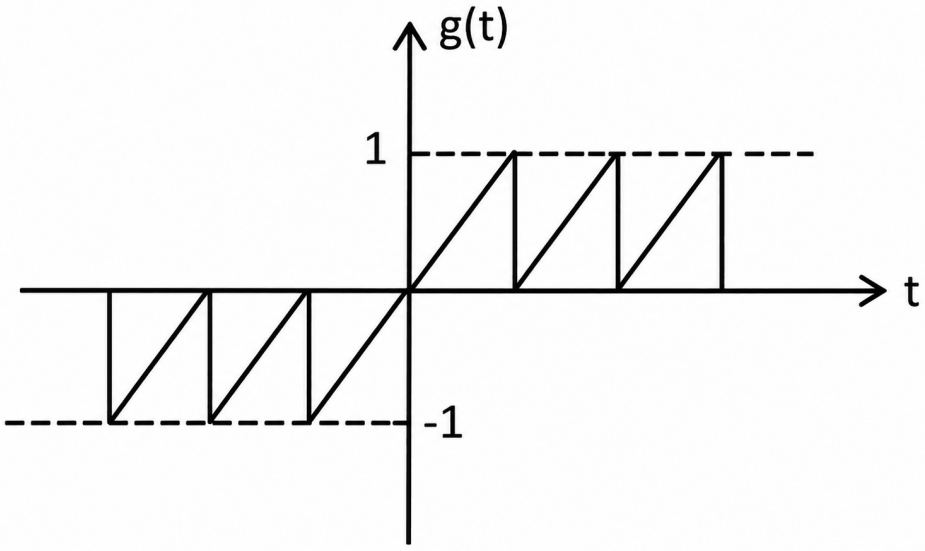


Fig. Solution Q3.17.1

For example, $g(-1/2) = -1/2$ and $g(1/2) = 1/2$, so a one-unit shift fails.

More generally, take any proposed positive period T. Choose a small positive noninteger $\varepsilon < T$ for which $T - \varepsilon$ is not an integer. Then $g(-\varepsilon) < 0$ while $g(T - \varepsilon) > 0$, so the two values cannot be equal. No positive period exists.
Therefore a Fourier series and its second harmonic coefficient are undefined. The source question is marked MTA (marks to all); that is an exam disposition, not a numerical answer.

Q3.16.1 MCQ | 2016 | 1M | Ans: B | ● ○ ○

Given,

$$f(-x) = -f(x),$$

which means $f(x)$ is an **odd** function.
The general Fourier series is

$$f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos(kx) + \sum_{k=1}^{\infty} b_k \sin(kx).$$

Since $f(x)$ is odd,

$$a_0 = 0, \qquad a_k = 0, \ \forall k.$$

Hence, only sine terms remain in the Fourier series:

$$f(x) = \sum_{k=1}^{\infty} b_k \sin(kx).$$

There is no restriction that only odd harmonics are present. Therefore, all sine harmonics may exist.

$$\text{(b) } f(x) = \sum_{k=1}^{\infty} b_k \sin(kx)$$

Key Points

- An odd function satisfies $f(-x) = -f(x)$.
- The Fourier series of an odd function contains only sine terms.
- Even symmetry removes sine terms, while odd symmetry removes cosine terms and the DC component.

Q3.16.2 MCQ | 2016 | 2M | Ans: D | ● ○ ○

For a continuous-time LTI system, the complex exponential $e^{j\omega t}$ is an Eigen-signal. Since the impulse response $h(t)$ is **real and even**,

$$T\{e^{j\omega t}\} = H(j\omega)e^{j\omega t},$$

where

$$H(j\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t}\, dt.$$

Because $h(t)$ is real and even,

$$H(j\omega) = H(-j\omega),$$

and the frequency response is purely real.
Now,

$$\cos t = \frac{e^{jt} + e^{-jt}}{2}.$$

Applying the system,

$$T\{\cos t\} = \frac{H(j)e^{jt} + H(-j)e^{-jt}}{2} = H(j)\cos t.$$

Similarly,

$$\sin t = \frac{e^{jt} - e^{-jt}}{2j},$$

hence

$$T\{\sin t\} = \frac{H(j)e^{jt} - H(-j)e^{-jt}}{2j} = H(j)\sin t.$$

Thus, both $\cos t$ and $\sin t$ remain unchanged except for multiplication by the same Eigenvalue $H(j)$.

(d) $\cos(t)$ and $\sin(t)$ are both Eigen-signals with identical Eigenvalues.

Q3.15.1 MCQ 2015 2M Ans: D ● ○ ○

Given,

$$\text{sgn}(x) = \begin{cases} \frac{x}{|x|}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

The signal is

$$f(t) = \text{sgn}(\cos t).$$

Since

$$\cos(-t) = \cos t,$$

we have

$$f(-t) = \text{sgn}(\cos(-t)) = \text{sgn}(\cos t) = f(t).$$

Hence, $f(t)$ is an **even** function, so its Fourier series contains only cosine terms.
Also,

$$f(t + \pi) = \text{sgn}(\cos(t + \pi)) = -\text{sgn}(\cos t) = -f(t).$$

Thus, the signal has **half-wave symmetry**, which eliminates all even harmonics.
Therefore, the Fourier series contains only odd cosine harmonics:

$$f(t) = \frac{4}{\pi}\left(\cos t - \frac{1}{3}\cos 3t + \frac{1}{5}\cos 5t - \cdots\right).$$

(d) Only cosine terms with odd numbered harmonics

Key Points

- Even symmetry $\Rightarrow$ only cosine coefficients are present.
- Half-wave symmetry $\left(f(t + \pi) = -f(t)\right)$ eliminates all even harmonics.
- The Fourier series of $\text{sgn}(\cos t)$ contains only odd cosine harmonics.

Q3.14.1 MCQ 2014 2M Ans: C ● ○ ○

A 10 kHz even-symmetric square wave contains only odd harmonics in its Fourier series:

$$f(t) = \frac{4A}{\pi}\left[\cos(\omega_0 t) - \frac{1}{3}\cos(3\omega_0 t) + \frac{1}{5}\cos(5\omega_0 t) - \cdots\right],$$

where

$$\omega_0 = 2\pi(10\,\text{kHz}).$$

The band-pass filter has

$$f_c = 30\,\text{kHz}, \qquad \text{BW} = 6\,\text{kHz}.$$

Hence, it passes frequencies from

$$27\,\text{kHz to } 33\,\text{kHz}.$$

The third harmonic of the square wave is

$$3 \times 10\,\text{kHz} = 30\,\text{kHz},$$

which lies exactly at the center frequency of the filter.
All other harmonics (10 kHz, 50 kHz, 70 kHz, . . .) are heavily attenuated.
Therefore, only the third harmonic appears at the output:

$$v_o(t) \approx \frac{4A}{3\pi}\cos(2\pi \cdot 30\,\text{kHz}\, t)\,.$$

Thus, the output is a nearly pure cosine wave at 30 kHz.

(c) A nearly perfect cosine wave at 30 kHz

Q3.14.2 MCQ 2014 1M Ans: C ● ○ ○

A periodic square wave satisfies the **Dirichlet conditions**:

- It is single-valued and finite.
- It has a finite number of discontinuities in one period.
- It has a finite number of maxima and minima in one period.

Hence, its Fourier series coefficients exist.
Further, by the Dirichlet convergence theorem, the Fourier series converges:

- to the signal value at all points of continuity,
- to the average of the left and right limits at points of discontinuity.

Therefore, the reconstruction converges at almost every point (i.e., all points except the jump discontinuities).

(c) The Fourier series coefficients exist and the reconstruction converges at most points.

Q3.13.1 MCQ 2013 1M Ans: A ● ○ ○

Given,

$$v(t) = 30\sin(100t) + 10\cos(300t) + 6\sin\left(500t + \frac{\pi}{4}\right).$$

The angular frequencies present in the signal are

$$100,\ 300,\ 500\ \text{rad/s}.$$

The fundamental angular frequency is the greatest common divisor (GCD) of all component frequencies:

$$\omega_0 = \gcd(100, 300, 500) = 100\ \text{rad/s}.$$

The given signal can therefore be written as

$$v(t) = 30\sin(\omega_0 t) + 10\cos(3\omega_0 t) + 6\sin\left(5\omega_0 t + \frac{\pi}{4}\right),$$

showing that all components are harmonics of 100 rad/s.

(a) 100 rad/s

Q3.11.1 MCQ 2011 1M Ans: D ● ○ ○

From the given waveform,

$$f(-t) = f(t),$$

hence the signal is an **even** function.
Since the signal is even,

$$b_n = 0, \qquad \forall n.$$

Also, the waveform has a non-zero average value, therefore

$$a_0 \neq 0.$$

If the waveform is shifted downward by its DC component,

$$\frac{f_{\max}}{2},$$

the resulting waveform possesses half-wave symmetry.
Hence, all even harmonics vanish and only odd cosine harmonics remain:

$$a_n \neq 0, \qquad n = 1, 3, 5, \ldots$$

(d) a_0 and $a_n,\ n = 1, 3, 5, \ldots, \infty$

Key Points

- Removing the DC component makes the waveform half-wave symmetric.
- Half-wave symmetry eliminates all even harmonics.
- Therefore, only odd cosine harmonics together with the DC component are present.

Mistakes to be avoided

- Do not conclude that all cosine harmonics are present merely because the waveform is even.
- Check whether the waveform becomes half-wave symmetric after removing its DC component.

Q3.10.1 MCQ 2010 1M Ans: A ● ○ ○

The given waveform is a square wave of amplitude ±1 with 50% duty cycle.
The waveform satisfies the half-wave symmetry condition

$$f\left(t + \frac{T}{2}\right) = -f(t).$$

A signal with half-wave symmetry contains only odd harmonics in its Fourier series. Hence,

$$a_n = b_n = 0, \qquad n = 2, 4, 6, \ldots$$

Therefore, the second harmonic coefficients are

$$a_2 = 0, \qquad b_2 = 0.$$

The amplitude of the second harmonic is

$$A_2 = \sqrt{a_2^2 + b_2^2} = 0.$$

(a) 0

Q3.09.1 MCQ | 2009 | 2M | Ans: C | ● ○ ○

Given Fourier series coefficients,

$$a_{-2} = 2 - j, \quad a_{-1} = 0.5 + j0.2, \quad a_0 = j2, \quad a_1 = 0.5 - j0.2, \quad a_2 = 2 + j, \quad a_k = 0, \ |k| > 2.$$

Since only finitely many Fourier series coefficients are non-zero, the signal is still periodic. Therefore,

$$\text{Energy of } x(t) = \infty,$$

so Option A is incorrect.
The average (DC) value is

$$a_0 = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t)\, dt = j2 \neq 0,$$

hence Option B is also incorrect.
The coefficients satisfy

$$a_k = a_{-k}^*, \qquad k \neq 0.$$

Thus, all non-DC harmonics combine to produce a purely real signal.
The only imaginary contribution comes from

$$a_0 = j2,$$

which is a constant (DC) term. Hence,

$$x(t) = x_r(t) + j2,$$

where $x_r(t)$ is real.
Therefore,

$$\Im\{x(t)\} = 2,$$

which is constant.

(c) The imaginary part of $x(t)$ is constant

Key Points

- A periodic signal has infinite energy but finite average power.
- The DC Fourier coefficient a_0 equals the average value of the signal.
- If $a_k = a_{-k}^*$ for $k \neq 0$, the non-DC harmonics contribute only to the real part of the signal.
- A purely imaginary a_0 contributes only a constant imaginary component.

Mistakes to be avoided

- Do not conclude finite energy from finitely many Fourier coefficients.
- Do not assume the average value is zero for every periodic signal.
- Do not infer symmetry of the real part without verifying all Fourier coefficients, including the DC term.

Q3.08.1 MCQ | 2008 | 2M | Ans: B | ● ○ ○

Let the Fourier series coefficients of $x(t)$ be a_k. Using the time-shifting property,

$$x(t - t_0) \iff a_k e^{-jk\omega_0 t_0},$$

$$x(t + t_0) \iff a_k e^{jk\omega_0 t_0},$$

where

$$\omega_0 = \frac{2\pi}{T}.$$

The Fourier series coefficients of

$$y(t) = x(t - t_0) + x(t + t_0)$$

are

$$b_k = a_k \left(e^{-jk\omega_0 t_0} + e^{jk\omega_0 t_0}\right) = 2a_k \cos(k\omega_0 t_0).$$

Given that

$$b_k = 0, \qquad \forall \text{ odd } k,$$

and a_k need not be zero,

$$\cos(k\omega_0 t_0) = 0, \qquad \forall \text{ odd } k.$$

Taking $k = 1$,

$$\omega_0 t_0 = \frac{\pi}{2}$$

(the smallest positive solution). Hence,

$$t_0 = \frac{\pi}{2\omega_0} = \frac{\pi}{2(2\pi/T)} = \frac{T}{4}.$$

For every odd k,

$$\cos\left(k\frac{\pi}{2}\right) = 0,$$

so the condition is satisfied.

$$\boxed{\text{(b) } t_0 = \frac{T}{4}}$$

Q3.07.1 MCQ | 2007 | 2M | Ans: A | ● ○ ○

The signal satisfies

$$x(t + T) = -x(t),$$

which means it has half-wave symmetry. Hence, the fundamental period is

$$T_0 = 2T, \qquad \omega_0 = \frac{2\pi}{T_0} = \frac{\pi}{T}.$$

Over one period,

$$x(t) = \begin{cases} 1, & -\frac{T}{4} < t \le \frac{3T}{4}, \\ -1, & \frac{3T}{4} < t \le \frac{7T}{4}. \end{cases}$$

The waveform is a 50% duty-cycle square wave centered at

$$t = \frac{T}{4}.$$

The fundamental component of a unit-amplitude square wave is

$$\frac{4}{\pi}\cos\left[\omega_0\left(t-\frac{T}{4}\right)\right].$$

Substituting

$$\omega_0 = \frac{\pi}{T},$$

gives

$$x_1(t) = \frac{4}{\pi}\cos\left(\frac{\pi t}{T}-\frac{\pi}{4}\right).$$

$$\boxed{\text{(a) } \frac{4}{\pi}\cos\left(\frac{\pi t}{T}-\frac{\pi}{4}\right)}$$

Q3.06.1 MCQ | 2006 | 1M | Ans: C | ● ○ ○

The trigonometric Fourier series of a real-valued signal is

$$x(t) = a_0 + \sum_{k=1}^{\infty}\left[a_k\cos(k\omega_0 t) + b_k\sin(k\omega_0 t)\right],$$

where

$$\omega_0 = \frac{2\pi}{T}.$$

Since no sine terms are present,

$$b_k = 0,$$

hence

$$x(t) = x(-t),$$

i.e., the signal is even.
Also, there are no even harmonics, so

$$x(t) = a_0 + \sum_{\substack{k=1\\ k\text{ odd}}}^{\infty} a_k\cos(k\omega_0 t).$$

Now,

$$x\left(t-\frac{T}{2}\right) = a_0 + \sum_{\text{odd}} a_k\cos(k\omega_0 t - k\pi).$$

Since

$$\cos(k\pi) = -1, \qquad k \text{ odd},$$

we obtain

$$x\left(t-\frac{T}{2}\right) = a_0 - \sum_{\text{odd}} a_k\cos(k\omega_0 t).$$

As the listed options imply zero DC component ($a_0 = 0$),

$$x\left(t-\frac{T}{2}\right)=-x(t).$$

Using periodicity,

$$x(T-t)=x(-t)=x(t),$$

and therefore,

$$-x\left(-t-\frac{T}{2}\right)=-\left[-x(-t)\right]=x(-t)=x(t).$$

Hence,

$$x(t)=x(T-t)=-x\left(-t-\frac{T}{2}\right).$$

$$\boxed{\text{(c) } x(t)=x(T-t)=-x\left(-t-\frac{T}{2}\right)}$$

Q3.05.1 MCQ | 2005 | 2M | Ans: D | ● ○ ○

Using the trigonometric identity,

$$\sin^2 x=\frac{1-\cos 2x}{2}.$$

Substituting the identity directly,

$$f(x)=\sin^2 x=\frac{1-\cos 2x}{2}=\frac{1}{2}-\frac{1}{2}\cos 2x.$$

This is already the Fourier series expansion of the given function.

$$\boxed{\text{(d) } 0.5-0.5\cos 2x}$$

Q3.02.1 MCQ | 2002 | 1M | Ans: C | ● ○ ○

The shown triangle is even, has period $T=2$, and satisfies $f(t)=1-2|t|$ on $[-1,1]$. Thus it has zero average and no sine coefficients. With $\omega_0=\pi$, integration gives

$$a_n=2\int_0^1(1-2t)\cos(n\pi t)\,dt$$

$$a_n=\frac{4[1-(-1)^n]}{n^2\pi^2}.$$

All odd cosine coefficients are positive; the even coefficients vanish. Therefore

$$f(t)=\frac{8}{\pi^2}\left[\cos(\pi t)+\frac{\cos(3\pi t)}{9}+\frac{\cos(5\pi t)}{25}+\cdots\right].$$

This is option $\boxed{\text{C}}$. Checking $t=0$ also confirms the all-positive series sums to the peak 1.

Q3.01.1 MCQ | 2001 | 5M | Ans: 0.2, 1.21, 0.98 | ● ● ●

(A) DC Component

$$V_{DC} = \frac{3(1) + 2(-1)}{5} = 0.2\,\text{V}.$$

$$\boxed{V_{DC} = 0.2\,\text{V}}$$

(B) The amplitude of the fundamental component of v

To find the amplitude of the fundamental component, we use the trigonometric Fourier series. The amplitude C_1 of the fundamental frequency is given by $C_1 = \sqrt{a_1^2 + b_1^2}$, where a_1 and b_1 are the first Fourier coefficients.

Step 1: Calculate a_1

$$a_1 = \frac{2}{T}\int_0^T v(t)\cos(\omega_0 t)\,dt$$

$$a_1 = \frac{2}{5}\left[\int_0^3 (1)\cos\left(\frac{2\pi}{5}t\right)dt + \int_3^5 (-1)\cos\left(\frac{2\pi}{5}t\right)dt\right]$$

$$a_1 = \frac{2}{5}\left[\frac{5}{2\pi}\sin\left(\frac{2\pi}{5}t\right)\Big|_0^3 - \frac{5}{2\pi}\sin\left(\frac{2\pi}{5}t\right)\Big|_3^5\right]$$

$$a_1 = \frac{1}{\pi}\left[\left(\sin\left(\frac{6\pi}{5}\right) - 0\right) - \left(\sin(2\pi) - \sin\left(\frac{6\pi}{5}\right)\right)\right]$$

Since $\sin(2\pi) = 0$:

$$a_1 = \frac{1}{\pi}\left[2\sin\left(\frac{6\pi}{5}\right)\right] = \frac{2}{\pi}\sin\left(\frac{6\pi}{5}\right)$$

Step 2: Calculate b_1

$$b_1 = \frac{2}{T}\int_0^T v(t)\sin(\omega_0 t)\,dt$$

$$b_1 = \frac{2}{5}\left[\int_0^3 (1)\sin\left(\frac{2\pi}{5}t\right)dt - \int_3^5 \sin\left(\frac{2\pi}{5}t\right)dt\right]$$

$$b_1 = \frac{2}{5}\left[-\frac{5}{2\pi}\cos\left(\frac{2\pi}{5}t\right)\Big|_0^3 + \frac{5}{2\pi}\cos\left(\frac{2\pi}{5}t\right)\Big|_3^5\right]$$

$$b_1 = \frac{1}{\pi}\left[\left(-\cos\left(\frac{6\pi}{5}\right) + \cos(0)\right) + \left(\cos(2\pi) - \cos\left(\frac{6\pi}{5}\right)\right)\right]$$

Since $\cos(0) = 1$ and $\cos(2\pi) = 1$:

$$b_1 = \frac{1}{\pi}\left[2 - 2\cos\left(\frac{6\pi}{5}\right)\right] = \frac{2}{\pi}\left[1 - \cos\left(\frac{6\pi}{5}\right)\right]$$

Step 3: Calculate Amplitude C_1

$$C_1 = \sqrt{a_1^2 + b_1^2}$$

$$C_1 = \sqrt{\left(\frac{2}{\pi}\sin\left(\frac{6\pi}{5}\right)\right)^2 + \left(\frac{2}{\pi}\left(1 - \cos\left(\frac{6\pi}{5}\right)\right)\right)^2}$$

$$C_1 = \frac{2}{\pi}\sqrt{\sin^2\left(\frac{6\pi}{5}\right) + 1 - 2\cos\left(\frac{6\pi}{5}\right) + \cos^2\left(\frac{6\pi}{5}\right)}$$

Using the identity $\sin^2(\theta) + \cos^2(\theta) = 1$:

$$C_1 = \frac{2}{\pi}\sqrt{2 - 2\cos\left(\frac{6\pi}{5}\right)} = \frac{2\sqrt{2}}{\pi}\sqrt{1 - \cos\left(\frac{6\pi}{5}\right)}$$

Using the half-angle formula $1 - \cos(\theta) = 2\sin^2\left(\frac{\theta}{2}\right)$:

$$C_1 = \frac{2\sqrt{2}}{\pi}\sqrt{2\sin^2\left(\frac{3\pi}{5}\right)} = \frac{4}{\pi}\sin\left(\frac{3\pi}{5}\right)$$

The amplitude of the fundamental component is $\frac{4}{\pi}\sin\left(\frac{3\pi}{5}\right)$ V (approximately 1.21 V)

(C) The RMS value of the ac part of v

The total RMS value of a waveform squared is equal to the sum of the DC component squared and the AC RMS component squared:

$$V_{rms}^2 = V_{dc}^2 + V_{ac,rms}^2$$

Step 1: Find the total RMS value (V_{rms})

Since the waveform simply alternates between 1 V and −1 V, the squared waveform $v^2(t)$ is constantly 1 V^2 for the entire period.

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt} = \sqrt{\frac{1}{5}\int_0^5 (1)^2\,dt} = \sqrt{\frac{1}{5}[5]} = 1 \text{ V}$$

Step 2: Solve for $V_{ac,rms}$

Using the values we found earlier ($V_{rms} = 1$ V and $V_{dc} = 0.2$ V):

$$(1)^2 = (0.2)^2 + V_{ac,rms}^2$$

$$1 = 0.04 + V_{ac,rms}^2$$

$$V_{ac,rms}^2 = 0.96$$

$$V_{ac,rms} = \sqrt{0.96} = \sqrt{\frac{96}{100}} = \frac{4\sqrt{6}}{10} = \frac{2\sqrt{6}}{5}$$

The RMS value of the ac part is $\sqrt{0.96}$ V or $\frac{2\sqrt{6}}{5}V$ (approximately 0.98 V)

Q3.00.1 MCQ | 2000 | 2M | Ans: D | ● ○ ○

An AC waveform containing harmonics can be expressed as

$$v(t) = V_1 \sin(\omega t) + V_2 \sin(2\omega t + \phi_2) + V_3 \sin(3\omega t + \phi_3) + \cdots .$$

The addition of harmonic components changes the shape of the waveform. Consequently,

- The **peak value** changes because the harmonics may add constructively or destructively.
- The **RMS value** changes since the harmonic components contribute additional power.
- The **average value** over a half-cycle (rectified average) also changes because the waveform shape is no longer purely sinusoidal.

Hence, the distorted waveform differs from its fundamental component in all these measures.

(d) All the three measures (peak, RMS and average values)

Q3.97.1 MCQ | 1997 | 5M | Ans: 1.685 | ● ○ ○

Over one period $(0 \le t < 2\pi)$, the waveform is

$$x(t) = \begin{cases} 1, & 0 \le t < \dfrac{\pi}{3}, \\ 2, & \dfrac{\pi}{3} \le t < \dfrac{2\pi}{3}, \\ 0, & \dfrac{2\pi}{3} \le t < \pi, \\ -1, & \pi \le t < \dfrac{4\pi}{3}, \\ -2, & \dfrac{4\pi}{3} \le t < \dfrac{5\pi}{3}, \\ 0, & \dfrac{5\pi}{3} \le t < 2\pi. \end{cases}$$

The amplitude of the fundamental component is

$$C_1 = \sqrt{a_1^2 + b_1^2},$$

where

$$a_1 = \frac{1}{\pi}\int_0^{2\pi} x(t)\cos t\, dt, \qquad b_1 = \frac{1}{\pi}\int_0^{2\pi} x(t)\sin t\, dt.$$

Evaluating over the six intervals,

$$a_1 = \frac{\sqrt{3}}{\pi}, \qquad b_1 = \frac{5}{\pi}.$$

Hence,

$$C_1 = \sqrt{\left(\frac{\sqrt{3}}{\pi}\right)^2 + \left(\frac{5}{\pi}\right)^2} = \frac{\sqrt{28}}{\pi} = \frac{2\sqrt{7}}{\pi}.$$

Therefore, the amplitude of the fundamental component is

$$\boxed{\frac{2\sqrt{7}}{\pi} = 1.685}.$$

Q3.96.1 MCQ | 1996 | 1M | Ans: D

From the waveform, one complete cycle consists of

$$+V \text{ for } 2\,\text{ms}, \qquad -V \text{ for } 2\,\text{ms}.$$

Hence, the time period is

$$T = 4\,\text{ms} = 4 \times 10^{-3}\,\text{s}.$$

The fundamental frequency is

$$f_0 = \frac{1}{T} = \frac{1}{4 \times 10^{-3}} = 250\,\text{Hz}.$$

The frequency of the fifth harmonic is

$$f_5 = 5f_0 = 5 \times 250 = 1250\,\text{Hz}.$$

(d) 1250 Hz

PrepFusion

SOLUTIONS

IV

Continuous Time Fourier Transform

Topics

1. Introduction to Fourier Transform
2. Properties of Fourier Transform
3. Fourier Transform of Standard Signals
4. Duality Property
5. FT of Rectangular, Triangular and Sampling Signals
6. Fourier Transform of Periodic Signals
7. Differentiation, Integration, Multiplication and Convolution Property
8. Parseval's Theorem
9. LTI Systems and Analog Filters
10. Nyquist Rate - Baseband and Bandpass Signals
11. Sampling Theorem and Applications

Unit Snapshot*

Stream: EE	**Questions:** 32	**MCQ:** 25	**MSQ:** 0	**NAT:** 7
1M: 12	**2M:** 20	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2024)–4M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q4.25.1 NAT | 2025 | 2M | Ans: 200 | ● ○ ○

For an ideal low-pass filter,

$$H(j\omega) = \begin{cases} 1, & |\omega| < 200\pi, \\ 0, & \text{otherwise.} \end{cases}$$

Using the inverse Fourier transform,

$$h(t) = \frac{1}{2\pi} \int_{-200\pi}^{200\pi} e^{j\omega t}\, d\omega = \frac{\sin(200\pi t)}{\pi t}.$$

At $t = 0$, use the standard limit

$$\lim_{t \to 0} \frac{\sin(at)}{t} = a.$$

Hence,

$$h(0) = \frac{200\pi}{\pi} = 200.$$

$$\boxed{200}$$

Key Points

- For an ideal LPF with cutoff frequency ω_c, $h(t) = \dfrac{\sin(\omega_c t)}{\pi t}$.
- The value at the origin is $h(0) = \dfrac{\omega_c}{\pi}$.

Q4.24.1 NAT | 2024 | 1M | Ans: 0 | ● ○ ○

Using the Fourier transform property,

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\, dt,$$

$$\frac{dX(\omega)}{d\omega} = -j \int_{-\infty}^{\infty} t\, x(t) e^{-j\omega t}\, dt.$$

At $\omega = 0$,

$$\left.\frac{dX(\omega)}{d\omega}\right|_{\omega=0} = -j \int_{-\infty}^{\infty} t\, x(t)\, dt.$$

Given

$$x(t) = e^{-t^4} \cos t,$$

both e^{-t^4} and $\cos t$ are even functions. Hence $x(t)$ is even, so

$$t\, x(t)$$

is an odd function. Therefore,

$$\int_{-\infty}^{\infty} t\, x(t)\, dt = 0.$$

Thus,

$$\left.\frac{dX(\omega)}{d\omega}\right|_{\omega=0} = 0.$$

$$\boxed{0}$$

Q4.23.1 MCQ | 2023 | 1M | Ans: A | ● ○ ○

Using the inverse Fourier transform,

$$x(t) = \frac{1}{2\pi}\int_{-W_0}^{W_0} e^{j\omega t}\, d\omega = \frac{\sin(W_0 t)}{\pi t}.$$

$$x(0) = \lim_{t\to 0}\frac{\sin(W_0 t)}{\pi t} = \frac{W_0}{\pi}.$$

Hence, $x(0)$ increases with W_0, so option B is false.
Also, as $W_0 \to \infty$,

$$x(t) = \frac{\sin(W_0 t)}{\pi t} \longrightarrow \delta(t),$$

therefore option A is true.
Now, at

$$t = \frac{\pi}{2W_0},$$

$$x\left(\frac{\pi}{2W_0}\right) = \frac{\sin(\pi/2)}{\pi\left(\frac{\pi}{2W_0}\right)} = \frac{2W_0}{\pi^2} \neq \pm\frac{1}{\pi}.$$

Hence, both options C and D are false.
Therefore, the correct statement is

(a) $x(t)$ tends to be an impulse as $W_0 \to \infty$.

Q4.23.2 NAT | 2023 | 2M | Ans: 80 | ● ○ ○

Using the trigonometric identity,

$$2\cos A \cos B = \cos(A+B) + \cos(A-B),$$

the given signal becomes

$$x(t) = \cos(240\pi t) + \cos(120\pi t).$$

Hence, the original frequency components are

$$f_1 = \frac{240\pi}{2\pi} = 120\,\text{Hz}, \qquad f_2 = \frac{120\pi}{2\pi} = 60\,\text{Hz}.$$

The sampling frequency is

$$f_s = 200\,\text{Hz}, \qquad \frac{f_s}{2} = 100\,\text{Hz}.$$

The 60 Hz component lies below the Nyquist frequency and remains unchanged.
The 120 Hz component aliases to

$$f_a = |f_s - 120| = |200 - 120| = 80\,\text{Hz}.$$

Therefore, after sampling, the spectrum contains frequency components at

60 Hz and 80 Hz.

An ideal low-pass filter with cutoff frequency 100 Hz passes both components.
Hence, the maximum frequency present in the filtered signal is

80

Q4.22.1 MCQ 2022 2M Ans: C ● ○ ○

The impulse response is

$$h(t) = 2\left(\frac{\sin(10\pi t)}{\pi t}\right)\cos(40\pi t).$$

Using the Fourier transform pair,

$$2\frac{\sin(10\pi t)}{\pi t} \iff \begin{cases} 2, & |\omega| < 10\pi, \\ 0, & \text{otherwise.} \end{cases}$$

Multiplication by $\cos(40\pi t)$ shifts the spectrum to

$$30\pi < |\omega| < 50\pi.$$

The input signal contains frequency components at

$$10\pi,\ 15\pi,\ 42\pi,\ 45\pi.$$

Only the components at 42π and 45π lie within the passband. Hence,

$$y(t) = 7\sin(42\pi t) + 4\cos(45\pi t).$$

$$\boxed{\text{(c) } 7\sin(42\pi t) + 4\cos(45\pi t)}$$

Q4.21.1 NAT 2021 2M Ans: 1 ● ○ ○

The given signal is

$$x(t) = \begin{cases} 1 - |t|, & |t| \le 1, \\ 0, & |t| > 1. \end{cases}$$

Since $x(t)$ is real and non-negative,

$$|X(\omega)| \le X(0).$$

Hence, the maximum magnitude of $X(\omega)$ occurs at $\omega = 0$.
Now,

$$X(0) = \int_{-\infty}^{\infty} x(t)\,dt = \int_{-1}^{1} (1 - |t|)\,dt.$$

Using symmetry,

$$X(0) = 2\int_{0}^{1} (1 - t)\,dt = 2\left[t - \frac{t^2}{2}\right]_0^1 = 2\left(1 - \frac{1}{2}\right) = 1.$$

Therefore,

$$\boxed{1}$$

Q4.21.2 MCQ 2021 2M Ans: A ● ○ ○

Given,

$$\frac{dF(\omega)}{d\omega} = -\omega F(\omega), \qquad F(0) = 1.$$

Separating variables,

$$\frac{dF}{F} = -\omega\,d\omega.$$

Integrating,

$$\ln F(\omega) = -\frac{\omega^2}{2} + C.$$

Using $F(0) = 1$,

$$C = 0,$$

hence

$$F(\omega) = e^{-\omega^2/2}.$$

Using the inverse Fourier transform,

$$f(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)\, d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-\omega^2/2}\, d\omega.$$

Since

$$\int_{-\infty}^{\infty} e^{-\omega^2/2}\, d\omega = \sqrt{2\pi},$$

we obtain

$$f(0) = \frac{\sqrt{2\pi}}{2\pi} = \frac{1}{\sqrt{2\pi}} \approx 0.399 < 1.$$

Therefore,

(a) $f(0) < 1$

Q4.20.1 MCQ | 2020 | 2M | Ans: B

The poles are

$$(2, -1) = 2 - j, \qquad (-2, 1) = -2 + j,$$

and the zeros are

$$(2, 1) = 2 + j, \qquad (-2, -1) = -2 - j.$$

$$H(s) = K\frac{(s - (2 + j))(s - (-2 - j))}{(s - (2 - j))(s - (-2 + j))}.$$

For a causal realization, the ROC lies to the right of the rightmost pole. Since one pole is at

$$s = 2 - j,$$

the ROC is

$$\Re\{s\} > 2,$$

which does not include the $j\omega$-axis. Hence, the system is **unstable**.
Also, the poles and zeros are mirror images about the imaginary axis:

$$2 - j \longleftrightarrow -2 - j, \qquad -2 + j \longleftrightarrow 2 + j.$$

Therefore,

$$|H(j\omega)| = 1,$$

so the system is an **all-pass** filter.
Since the poles and zeros are not in complex-conjugate pairs, the coefficients of $H(s)$ are **complex**.

(b) Unstable, complex, All pass

Q4.18.1 NAT 2018 2M Ans: 10 ● ○ ○

Given,

$$X(\omega) = \frac{1}{(10 + j\omega)^2}.$$

Using the Fourier transform pair,

$$e^{-at}u(t) \iff \frac{1}{a + j\omega},$$

and the differentiation property,

$$te^{-at}u(t) \iff \frac{1}{(a + j\omega)^2}.$$

Hence,

$$x(t) = te^{-10t}u(t).$$

At $t = 1$,

$$x(1) = e^{-10}.$$

Therefore,

$$|\ln x(1)| = |\ln(e^{-10})| = |-10| = 10.$$

Thus,

$$\boxed{10.0}$$

Q4.17.1 MCQ 2017 2M Ans: C ● ○ ○

The impulse train has impulses spaced by

$$\frac{1}{2000}\ \text{s},$$

with alternating signs. Hence, the signal is periodic with

$$T = \frac{1}{1000}\ \text{s}, \qquad \omega_0 = \frac{2\pi}{T} = 2000\pi\ \text{rad/s}.$$

The Fourier series coefficients are

$$a_k = \frac{1}{T}\left[1 - e^{-jk\pi}\right] = 1000\left[1 - (-1)^k\right].$$

Thus,

$$a_k = \begin{cases} 2000, & k \text{ odd}, \\ 0, & k \text{ even}. \end{cases}$$

The filter has

$$H(\omega) = \begin{cases} 1, & |\omega| < 5000\pi, \\ 0, & \text{otherwise}. \end{cases}$$

Hence only the harmonics at

$$\omega = \pm 2000\pi$$

are passed. The harmonics at 0 and $\pm 4000\pi$ have zero coefficients, while $\pm 6000\pi$ are rejected.
Therefore,

$$y(t) = 2000e^{j2000\pi t} + 2000e^{-j2000\pi t} = 4000\cos(2000\pi t).$$

$$\boxed{\text{(c) } 4000\cos(2000\pi t)}$$

Q4.17.2 MCQ 2017 1M Ans: A ● ○ ○

Given,

$$z(t) = x(t) * y(t).$$

Then,

$$z(ct) = \int_{-\infty}^{\infty} x(\tau)\, y(ct - \tau)\, d\tau.$$

Let

$$\tau = c\lambda, \qquad d\tau = c\, d\lambda.$$

Hence,

$$z(ct) = c \int_{-\infty}^{\infty} x(c\lambda)\, y(c(t - \lambda))\, d\lambda.$$

Recognizing the convolution of the scaled signals,

$$z(ct) = c\,[x(ct) * y(ct)].$$

Therefore,

$$\boxed{\text{(a) } c \cdot x(ct) * y(ct)}$$

Q4.17.3 MCQ 2017 2M Ans: B ● ○ ○

The given input spectrum is bandlimited to

$$|\omega| \le 1000\pi \;\Rightarrow\; |f| \le 500\,\text{Hz}.$$

Multiplication by

$$\cos(1000\pi t) = \cos(2\pi \cdot 500\, t)$$

shifts the spectrum by ±500 Hz:

$$Y_1(f) = \frac{1}{2}\Big[X(f - 500) + X(f + 500)\Big].$$

Hence, the spectrum occupies

$$[-1000,\, 0]\ \text{Hz} \quad \text{and} \quad [0,\, 1000]\ \text{Hz},$$

i.e.,

$$|f| \le 1000\,\text{Hz}.$$

The impulse response

$$h(t) = \frac{\sin(1500\pi t)}{\pi t}$$

is an ideal low-pass filter with cutoff frequency

$$f_c = \frac{1500\pi}{2\pi} = 750\,\text{Hz}.$$

Therefore, after filtering,

$$|f| \le 750\,\text{Hz}.$$

Using the Nyquist sampling criterion,

$$f_s^{\min} = 2f_{\max} = 2 \times 750 = 1500\ \text{samples/s}.$$

$$\boxed{\text{(b) 1500 samples/s}}$$

Q4.16.1 MCQ 2016 2M Ans: D ●○○

The original GATE 2016 EE Set 2 Q29 asks for the integral of the signal, not its square. For $a > 0$, the Dirichlet integral is

$$\int_{-\infty}^{\infty} \frac{\sin(at)}{\pi t}\, dt = 1.$$

With $a = 2\pi$, the requested leading factor 2 therefore gives $\boxed{2}$, option (D).
The integral is an improper oscillatory integral, not an energy integral.

Q4.16.2 NAT 2016 1M Ans: C ●○○

The given signal is

$$x(t)\cos(2000\pi t).$$

Since

$$2000\pi = 2\pi f_c,$$

the carrier frequency is

$$f_c = \frac{2000\pi}{2\pi} = 1000\,\text{Hz} = 1\,\text{kHz}.$$

Multiplication by a cosine shifts the spectrum to

$$\pm f_c.$$

If the maximum frequency in $x(t)$ is

$$f_m = 5\,\text{kHz},$$

then the shifted spectrum extends up to

$$f_{\max} = f_m + f_c = 5 + 1 = 6\,\text{kHz}.$$

Hence,

$$\boxed{6}$$

Q4.16.3 MCQ 2016 2M Ans: C ●○○

From the plots,

$$X_2(j\omega) = X_1(-j\omega).$$

Also, both $X_1(j\omega)$ and $X_2(j\omega)$ are purely real but are *not* even functions.
Using the Fourier transform properties,

$$X(\omega)\ \text{real} \iff x(t) = x^*(-t),$$

and

$$X(-\omega) \iff x(-t).$$

Hence,

$$x_2(t) = x_1(-t) = x_1^*(t).$$

Therefore,

$$x_1(t)x_2(t) = x_1(t)x_1^*(t) = |x_1(t)|^2,$$

which is always a real-valued signal.
Since $X_1(j\omega)$ is not Hermitian symmetric, $x_1(t)$ is complex. Similarly, $x_2(t)$ is also complex.
Thus,

$$\boxed{\text{(c) } x_1(t) \text{ and } x_2(t) \text{ are complex but } x_1(t)x_2(t) \text{ is real.}}$$

Key Points

- $X(\omega)$ real $\Longleftrightarrow x(t) = x^*(-t)$.
- Frequency reversal gives time reversal:

$$X(-\omega) \Longleftrightarrow x(-t).$$

Q4.16.4 MCQ | 2016 | 2M | Ans: B | ● ○ ○

The output of the multiplier is

$$v(t) = x_1(t)x_2(t).$$

Using the multiplication property of the Fourier Transform,

$$V(\omega) = \frac{1}{2\pi} X_1(\omega) * X_2(\omega),$$

where $*$ denotes convolution.
Since $X_1(\omega)$ is nonzero over

$$-B_1 \le \omega \le B_1,$$

and $X_2(\omega)$ is nonzero over

$$-B_2 \le \omega \le B_2,$$

their convolution is nonzero over

$$-(B_1 + B_2) \le \omega \le (B_1 + B_2).$$

The LTI system has

$$h(t) = e^{-2|t|},$$

whose frequency response is

$$H(\omega) = \frac{4}{\omega^2 + 4}.$$

Since $H(\omega) \neq 0$ for all finite ω, the bandwidth of the signal remains

$$\omega_{\max} = B_1 + B_2.$$

Applying the Nyquist sampling criterion,

$$\omega_{s,\min} = 2(B_1 + B_2).$$

Hence, the minimum sampling rate required is

$$2(B_1 + B_2).$$

$$\boxed{\text{(b) } 2(B_1 + B_2)}$$

Key Points

- Multiplication in time corresponds to convolution in the frequency domain.
- The support of the convolution equals the sum of the individual bandwidths.
- An LTI system with nonzero frequency response does not change the signal bandwidth.
- Nyquist rate is twice the highest frequency present.

Mistakes to be avoided

- Do not add the bandwidths after filtering unless the filter truncates the spectrum.
- Do not confuse multiplication in time with multiplication in frequency.

Q4.15.1 MCQ 2015 1M Ans: B ●○○

Filter F_1 is a low-pass filter with cut-off frequency f_1, while F_2 is a high-pass filter with cut-off frequency f_2.
The op-amp is a differential amplifier with equal input resistors R and feedback resistor $R/2$. Hence,

$$v_o = \frac{R/2}{R}(v_{F2} - v_{F1}) = \frac{1}{2}(v_{F2} - v_{F1}).$$

Now consider different frequency ranges:

- For $f < f_1$: F_1 passes and F_2 blocks, so $v_o \neq 0$.
- For $f_1 < f < f_2$: both filters attenuate the signal, so $v_o \approx 0$.
- For $f > f_2$: F_2 passes and F_1 blocks, so $v_o \neq 0$.

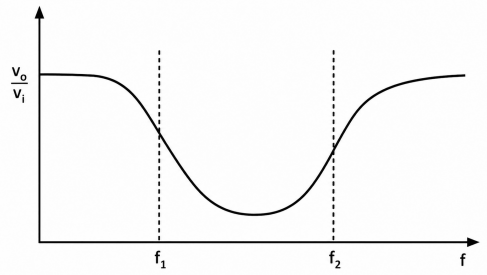


Fig. Solution Q4.15.1

Thus, the circuit attenuates only the middle band and passes both low and high frequencies. Therefore, it behaves as a band-stop (notch) filter.

(b) Band-stop filter

Q4.15.2 MCQ 2015 2M Ans: A ●○○

Using the Fourier transform definition,

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt.$$

Given,

$$x(t) = \begin{cases} e^{j10t}, & |t| \le 1, \\ 0, & |t| > 1. \end{cases}$$

Hence,

$$X(\omega) = \int_{-1}^{1} e^{j10t}e^{-j\omega t}\,dt = \int_{-1}^{1} e^{-j(\omega-10)t}\,dt.$$

Using

$$\int_{-T}^{T} e^{-jat}\,dt = \frac{2\sin(aT)}{a},$$

with $T = 1$ and $a = \omega - 10$,

$$X(\omega) = \frac{2\sin(\omega - 10)}{\omega - 10}.$$

Therefore,

(a) $\dfrac{2\sin(\omega - 10)}{\omega - 10}$

Q4.14.1 MCQ 2014 2M Ans: B ● ○ ○

Given $x(t)$ is a differentiable, non-constant, even function and

$$y(t) = \frac{dx(t)}{dt}.$$

Since the derivative of an even function is odd,

$$x(t) \text{ is even}, \qquad y(t) \text{ is odd}.$$

Using Fourier transform symmetry:

$$\text{Real even signal} \Longleftrightarrow \text{Real even spectrum}.$$

Hence,

$$X(\omega) \text{ is real}.$$

Also,

$$\text{Real odd signal} \Longleftrightarrow \text{Purely imaginary odd spectrum}.$$

Therefore,

$$Y(\omega) \text{ is imaginary}.$$

This is also verified using the differentiation property:

$$Y(\omega) = j\omega X(\omega).$$

Since $X(\omega)$ is real and $j\omega$ is purely imaginary,

$$Y(\omega)$$

is purely imaginary.

$$\boxed{\text{(b) } X(\omega) \text{ is real and } Y(\omega) \text{ is imaginary}}$$

Q4.14.2 MCQ 2014 2M Ans: D ● ○ ○

From the pole-zero plot, all poles and zeros lie on the same circle and are symmetric about the imaginary axis. For any point on the $j\omega$-axis,

$$|H(j\omega)| = K\frac{\prod_i |j\omega - z_i|}{\prod_k |j\omega - p_k|}.$$

Since every pole has a corresponding zero at the mirror location about the imaginary axis and all poles and zeros have the same magnitude, the pole-zero distance ratios cancel each other for every value of ω.
Hence,

$$|H(j\omega)| = \text{constant}, \qquad -\infty < \omega < \infty.$$

Thus, the given system is an all-pass filter.

$$\boxed{\text{(d) } |H(\omega)| = \text{constant};\ -\infty < \omega < \infty}$$

Q4.14.3 MCQ 2014 1M Ans: A ● ○ ○

Given,

$$x(t) = \begin{cases} 1, & |t| < 1, \\ 0, & |t| > 1. \end{cases}$$

Its Fourier transform is

$$X(\omega) = \int_{-1}^{1} e^{-j\omega t}\, dt = \frac{2\sin\omega}{\omega}.$$

Using the time-scaling property,

$$x(at) \Longleftrightarrow \frac{1}{|a|} X\left(\frac{\omega}{a}\right),$$

we get

$$x(2t) \Longleftrightarrow \frac{1}{2} X\left(\frac{\omega}{2}\right) = \frac{2\sin(\omega/2)}{\omega},$$

and

$$x\left(\frac{t}{2}\right) \Longleftrightarrow 2X(2\omega) = \frac{2\sin(2\omega)}{\omega}.$$

Since

$$y(t) = x(2t) * x\left(\frac{t}{2}\right),$$

the convolution property gives

$$Y(\omega) = \frac{2\sin(\omega/2)}{\omega} \cdot \frac{2\sin(2\omega)}{\omega} = \frac{4}{\omega^2} \sin\left(\frac{\omega}{2}\right) \sin(2\omega).$$

Therefore,

$$\boxed{\text{(a) } \frac{4}{\omega^2} \sin\left(\frac{\omega}{2}\right) \sin(2\omega)}$$

Q4.14.4 MCQ 2014 1M Ans: C ● ○ ○

From the figure,

$$f(-t) = -f(t),$$

hence $f(t)$ is a real-valued odd function.
For Fourier transforms,

$$\text{Real odd signal} \Longleftrightarrow \text{Purely imaginary odd spectrum.}$$

Therefore,

$$F(\omega)$$

is purely imaginary and odd.

$$\boxed{\text{(c) Imaginary and odd function of } \omega}$$

Q4.14.5 MCQ 2014 1M Ans: B ● ○ ○

For a sinusoidal input, the steady-state output is obtained by evaluating the transfer function at

$$s = j\omega.$$

Given,

$$H(s) = \frac{1}{s(s+4)},$$

and the input is

$$\cos(3t),$$

so

$$\omega = 3 \text{ rad/s}.$$

Hence,

$$H(j3) = \frac{1}{(j3)(4+j3)}.$$

The output amplitude is

$$A = |H(j3)| = \frac{1}{|j3|\,|4+j3|} = \frac{1}{3 \times \sqrt{4^2+3^2}} = \frac{1}{3\times 5} = \frac{1}{15}.$$

Also,

$$\angle H(j3) = -\left(90^\circ + \tan^{-1}\frac{3}{4}\right) = -126.87^\circ.$$

Since

$$\cos(3t - 126.87^\circ) = \sin(3t - 36.87^\circ),$$

the output can be written in the form

$$A\sin(3t+\alpha),$$

where only the amplitude A is required.

$$\boxed{\text{(b) } \frac{1}{15}}$$

Q4.14.6 MCQ 2014 2M Ans: B ● ○ ○

The Fourier transform pair is

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}\,dt.$$

The inverse Fourier transform is

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{j\omega t}\,d\omega.$$

Given,

$$g(t) = \int_{-\infty}^{\infty} F(\omega)e^{-j\omega t}\,d\omega.$$

Replacing t by $-t$ in the inverse Fourier transform,

$$f(-t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{-j\omega t}\,d\omega.$$

Hence,

$$g(t) = 2\pi f(-t).$$

Therefore, $g(t)$ is proportional to $f(t)$ only when

$$f(-t) = f(t),$$

i.e., when $f(t)$ is an even function.

(b) $g(t)$ would be proportional to $f(t)$ if $f(t)$ is an even function

Q4.14.7 MCQ | 2014 | 2M | Ans: C | ● ○ ○

Let the sampling frequency be

$$f_s = \frac{1}{20\,\text{ms}} = 50\,\text{Hz}.$$

After sampling, the aliased frequency that appears within the ideal LPF passband is

$$f_a = |f - nf_s|,$$

where n is an integer.
Since the LPF cutoff frequency is 25 Hz and the output frequency is 20 Hz,

$$|f - 50n| = 20.$$

Checking the given options:

$$f = 10\,\text{Hz} \Rightarrow |10| = 10\,\text{Hz} \neq 20\,\text{Hz},$$
$$f = 30\,\text{Hz} \Rightarrow |30 - 50| = 20\,\text{Hz} \checkmark,$$
$$f = 60\,\text{Hz} \Rightarrow |60 - 50| = 10\,\text{Hz},$$
$$f = 90\,\text{Hz} \Rightarrow |90 - 100| = 10\,\text{Hz}.$$

Hence, the input frequency is 30 Hz.

(c) 30 Hz

Q4.14.8 NAT | 2014 | 1M | Ans: 14 | ● ○ ○

The given signal is

$$f(t) = 3\sin 8\pi t + 6\sin 12\pi t + \sin 14\pi t.$$

Comparing each term with the standard form $\sin(2\pi f t)$,

$$8\pi = 2\pi f_1 \Rightarrow f_1 = 4\,\text{Hz},$$
$$12\pi = 2\pi f_2 \Rightarrow f_2 = 6\,\text{Hz},$$
$$14\pi = 2\pi f_3 \Rightarrow f_3 = 7\,\text{Hz}.$$

The highest frequency component is

$$f_{\max} = 7\,\text{Hz}.$$

By the Nyquist sampling criterion,

$$f_s \geq 2f_{\max} = 2 \times 7 = 14\,\text{Hz}.$$

14

Q4.13.1 MCQ 2013 1M Ans: A ● ○ ○

A band-limited signal has the maximum frequency

$$f_{\max} = 5\,\text{kHz}.$$

According to the Nyquist sampling theorem, the minimum sampling frequency is

$$f_s \geq 2f_{\max} = 2 \times 5 = 10\,\text{kHz}.$$

Check each option against the Nyquist criterion:

$$\begin{aligned} 5\,\text{kHz} &< 10\,\text{kHz} \quad \text{(Not valid)},\\ 12\,\text{kHz} &> 10\,\text{kHz} \quad \text{(Valid)},\\ 15\,\text{kHz} &> 10\,\text{kHz} \quad \text{(Valid)},\\ 20\,\text{kHz} &> 10\,\text{kHz} \quad \text{(Valid)}. \end{aligned}$$

Hence, the only invalid sampling frequency is 5 kHz.

(a) 5 kHz

Q4.12.1 MCQ 2012 2M Ans: C ● ○ ○

Using the inverse Fourier transform,

$$h(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} H(j\omega)\,d\omega.$$

Given,

$$H(j\omega) = \frac{2\cos\omega\,\sin 2\omega}{\omega}.$$

Using the identity

$$2\sin A\cos B = \sin(A+B) + \sin(A-B),$$

we obtain

$$H(j\omega) = \frac{\sin 3\omega + \sin\omega}{\omega}.$$

Hence,

$$h(0) = \frac{1}{2\pi}\left[\int_{-\infty}^{\infty}\frac{\sin 3\omega}{\omega}\,d\omega + \int_{-\infty}^{\infty}\frac{\sin\omega}{\omega}\,d\omega\right].$$

Using the standard result

$$\int_{-\infty}^{\infty}\frac{\sin(a\omega)}{\omega}\,d\omega = \pi, \qquad a > 0,$$

we get

$$h(0) = \frac{1}{2\pi}(\pi + \pi) = 1.$$

(c) 1

Q4.10.1 MCQ 2010 2M Ans: D ● ○ ○

The given signal is

$$x(t) = \begin{cases} 1, & |t| \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Using Parseval's theorem,

$$\int_{-\infty}^{\infty} |x(t)|^2 \, dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 \, d\omega.$$

Now,

$$|x(t)|^2 = \begin{cases} 1, & |t| \le 1, \\ 0, & \text{otherwise,} \end{cases}$$

therefore,

$$\int_{-\infty}^{\infty} |x(t)|^2 \, dt = \int_{-1}^{1} 1 \, dt = 2.$$

Hence,

$$\int_{-\infty}^{\infty} |X(\omega)|^2 \, d\omega = 2\pi \times 2 = 4\pi.$$

Therefore,

(d) 4π

Q4.08.1 MCQ 2008 2M Ans: A ● ○ ○

The input and impulse response are

$$x(t) = \text{sinc}(\alpha t), \qquad h(t) = \text{sinc}(\beta t).$$

Using the Fourier transform pair,

$$\text{sinc}(at) \Longleftrightarrow \frac{1}{|a|} \text{rect}\left(\frac{\omega}{2\pi a}\right).$$

Hence,

$$X(\omega) = \frac{1}{|\alpha|} \text{rect}\left(\frac{\omega}{2\pi\alpha}\right),$$

$$H(\omega) = \frac{1}{|\beta|} \text{rect}\left(\frac{\omega}{2\pi\beta}\right).$$

The output spectrum is

$$Y(\omega) = X(\omega)H(\omega).$$

The product of two rectangular functions is another rectangular function whose bandwidth is the smaller of the two:

$$Y(\omega) = K \ \text{rect}\left(\frac{\omega}{2\pi \ \min(\alpha, \beta)}\right),$$

where K is a constant.
Taking the inverse Fourier transform,

$$y(t) = K \ \text{sinc}(\min(\alpha, \beta)t).$$

Therefore,

(a) K sinc(γt), $\gamma = \min(\alpha, \beta)$

Q4.07.1 MCQ 2007 1M Ans: D ● ○ ○

A sinusoidal input to a stable LTI system produces a sinusoid at the *same frequency*. Only its amplitude and phase may change.

$$x(t) = a_1 \sin(\omega_1 t + \phi_1)$$

The steady-state output is

$$y(t) = a_2 F(\omega_2 t + \phi_2),$$

where $F(\cdot)$ must itself be a sinusoidal function (either sine or cosine). Hence,

$$\omega_2 = \omega_1.$$

The amplitude and phase are determined by the system frequency response:

$$a_2 = a_1 |H(j\omega_1)|, \qquad \phi_2 = \phi_1 + \angle H(j\omega_1).$$

Thus, a_2 need not equal a_1, and ϕ_2 need not equal ϕ_1.

(d) F must be a "sine" or "cosine" function with $\omega_1 = \omega_2$.

Q4.07.2 MCQ 2007 1M Ans: B ● ○ ○

The given signal has the highest frequency

$$f_{\max} = 1\,\text{kHz}.$$

The sampling interval is

$$T_s = 1\,\text{ms}.$$

Hence, the sampling frequency is

$$f_s = \frac{1}{T_s} = \frac{1}{1 \times 10^{-3}} = 1\,\text{kHz}.$$

The Nyquist rate is

$$2f_{\max} = 2 \times 1 = 2\,\text{kHz}.$$

Since

$$f_s < 2f_{\max},$$

the Nyquist criterion is violated and aliasing occurs.
The spectrum of the sampled signal is

$$U_T(\omega) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} U(\omega - n\omega_s),$$

where

$$\omega_s = 2\pi f_s.$$

Because the replicas are spaced by only 1 kHz, which is less than the Nyquist rate, they overlap completely, producing a constant (flat) spectrum due to aliasing.

Option B

SOLUTIONS

V

Laplace Transform

Topics

1. Introduction to Laplace Transform
2. Laplace Transform of Standard Signals and Properties
3. Region of Convergence (ROC)
4. Inverse Laplace Transform
5. Interrelation Between EFSC, FT and LT
6. Eigen Values and Eigen Functions in LTI Systems
7. Unilateral Laplace Transform

Unit Snapshot*

Stream: EE	**Questions:** 22	**MCQ:** 21	**MSQ:** 0	**NAT:** 1
1M: 15	**2M:** 7	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2024)–4M(2010)

**See preface for how Avg Weightage and Range are calculated.*

Q5.24.1 MCQ 2024 1M Ans: A ● ○ ○

The given signal is

$$x(t) = e^{t^2}\,[u(t-1) - u(t-10)]\,.$$

Using the definition of the bilateral Laplace transform,

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}\,dt = \int_{1}^{10} e^{t^2}e^{-st}\,dt.$$

Since the signal is nonzero only over the finite interval

$$1 \le t < 10,$$

the integral is finite for every finite value of s. Hence, the Laplace transform converges for all values of $Re(s)$.

(a) $-\infty < Re(s) < \infty$

Q5.20.1 MCQ 2020 1M Ans: B ● ○ ○

Check each statement using the properties of the two-sided Laplace transform.

- **Option A:** False. The Laplace transform does not exist for every signal (e.g., e^{t^2}).
- **Option B:** True. A bounded signal that is non-zero only over a finite interval has a Laplace transform of the form

$$X(s) = \int_{t_1}^{t_2} x(t)e^{-st}\,dt,$$

  which is an entire function of s. Hence, it has **no finite poles**.
- **Option C:** False. The number of finite poles and finite zeros need not be equal.
- **Option D:** False. A weighted sum of shifted exponentials generally gives

$$X(s) = \sum_i \frac{A_i e^{-s\tau_i}}{s - a_i},$$

  which contains poles at $s = a_i$.

(b) It has no poles for any bounded signal that is non-zero only inside a finite time interval.

Q5.19.1 MCQ 2019 1M Ans: D ● ○ ○

Factorise the denominator and perform partial fraction expansion:

$$H(s) = \frac{s+3}{s^2+2s+1} = \frac{s+3}{(s+1)^2}.$$

Let

$$\frac{s+3}{(s+1)^2} = \frac{A}{s+1} + \frac{B}{(s+1)^2}.$$

Multiplying by $(s+1)^2$,

$$s + 3 = A(s+1) + B.$$

Comparing coefficients,

$$A = 1, \qquad A + B = 3 \;\Rightarrow\; B = 2.$$

Hence,

$$H(s) = \frac{1}{s+1} + \frac{2}{(s+1)^2}.$$

Using standard inverse Laplace transforms,

$$\mathcal{L}^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at},$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+a)^2}\right\} = te^{-at}.$$

Therefore,

$$h(t) = e^{-t} + 2te^{-t}, \qquad t \geq 0.$$

$$\boxed{\text{(d) } 2te^{-t} + e^{-t}}$$

Q5.19.2 MCQ 2019 1M Ans: D ●○○

Given

$$H(s) = \frac{a_1 s^2 + b_1 s + c_1}{a_2 s^2 + b_2 s + c_2},$$

with

$$a_1 = 0, \qquad b_1 = 0.$$

Hence,

$$H(s) = \frac{c_1}{a_2 s^2 + b_2 s + c_2}.$$

Evaluate the gain at the extreme frequencies:
At DC ($s = 0$),

$$H(0) = \frac{c_1}{c_2} \neq 0.$$

As frequency increases ($s = j\omega,\ \omega \to \infty$),

$$|H(j\omega)| \approx \frac{c_1}{a_2\omega^2} \longrightarrow 0.$$

Thus, the system passes low frequencies and attenuates high frequencies. Hence, it is a low-pass filter.

$$\boxed{\text{(d) Low pass filter}}$$

Key Points

- A low-pass filter has finite gain at $\omega = 0$ and zero gain as $\omega \to \infty$.
- Compare the highest powers of s in the numerator and denominator to determine the high-frequency behavior.
- A constant numerator indicates attenuation of high-frequency components.

Q5.16.1 MCQ 2016 1M Ans: D ●○○

Taking the Laplace transform of

$$\frac{dy(t)}{dt} + \frac{1}{6}y(t) = 3x(t),$$

with zero initial conditions,

$$\left(s + \frac{1}{6}\right)Y(s) = 3X(s).$$

Hence, the transfer function is

$$H(s) = \frac{Y(s)}{X(s)} = \frac{3}{s + \frac{1}{6}}.$$

Given

$$x(t) = 3e^{-t/3}u(t),$$

its Laplace transform is

$$X(s) = \frac{3}{s + \frac{1}{3}}.$$

Therefore,

$$Y(s) = H(s)X(s) = \frac{9}{\left(s + \frac{1}{6}\right)\left(s + \frac{1}{3}\right)}.$$

Using partial fractions,

$$Y(s) = \frac{54}{s + \frac{1}{6}} - \frac{54}{s + \frac{1}{3}}.$$

Taking the inverse Laplace transform,

$$y(t) = 54e^{-t/6}u(t) - 54e^{-t/3}u(t).$$

(d) $54e^{-t/6}u(t) - 54e^{-t/3}u(t)$

Q5.16.2 MCQ 2016 1M Ans: A ●○○

Use the frequency-shifting property of the Laplace transform.
The standard Laplace transform is

$$\mathcal{L}\{\sin(5t)u(t)\} = \frac{5}{s^2 + 25}.$$

Since

$$f(t) = e^{2t}\sin(5t)u(t),$$

apply the property

$$\mathcal{L}\{e^{at}f(t)\} = F(s - a).$$

Replacing s by $s - 2$,

$$F(s) = \frac{5}{(s-2)^2 + 25}.$$

Expanding the denominator,

$$(s-2)^2 + 25 = s^2 - 4s + 4 + 25 = s^2 - 4s + 29.$$

Hence,

$$F(s) = \frac{5}{s^2 - 4s + 29}.$$

(a) $\dfrac{5}{s^2 - 4s + 29}$

Q5.15.1 MCQ | 2015 | 1M | Ans: B | ● ○ ○

Given

$$\mathcal{L}\left\{2\sqrt{\frac{t}{\pi}}\right\} = s^{-3/2}.$$

Differentiate both sides with respect to s:

$$\mathcal{L}\{tf(t)\} = -\frac{dF(s)}{ds}.$$

Here,

$$f(t) = 2\sqrt{\frac{t}{\pi}}, \qquad F(s) = s^{-3/2}.$$

Hence,

$$-\frac{d}{ds}\left(s^{-3/2}\right) = \frac{3}{2}s^{-5/2}.$$

Also,

$$tf(t) = 2t\sqrt{\frac{t}{\pi}} = \frac{2}{\sqrt{\pi}}t^{3/2}.$$

Therefore,

$$\mathcal{L}\left\{\frac{2}{\sqrt{\pi}}t^{3/2}\right\} = \frac{3}{2}s^{-5/2}.$$

Since

$$g(t) = \sqrt{\frac{1}{\pi t}} = \frac{1}{\sqrt{\pi}}t^{-1/2},$$

we have

$$2t\,g(t) = 2\sqrt{\frac{t}{\pi}}.$$

Applying the time-multiplication property,

$$\mathcal{L}\{2t\,g(t)\} = -\frac{dG(s)}{ds} = s^{-3/2},$$

where $G(s) = \mathcal{L}\{g(t)\}$.
Thus,

$$\frac{dG(s)}{ds} = -s^{-3/2}.$$

Integrating,

$$G(s) = 2s^{-1/2}.$$

Hence,

$$\mathcal{L}\left\{\sqrt{\frac{1}{\pi t}}\right\} = s^{-1/2}.$$

$$\boxed{\text{(b) } s^{-1/2}}$$

Q5.14.1 MCQ 2014 1M Ans: B ● ○ ○

For the LTI system,

$$h(t) = e^{-5t}u(t),$$

the transfer function is

$$H(s) = \frac{1}{s+5}.$$

The output is

$$y(t) = e^{-3t}u(t) - e^{-5t}u(t),$$

whose Laplace transform is

$$Y(s) = \frac{1}{s+3} - \frac{1}{s+5} = \frac{2}{(s+3)(s+5)}.$$

Since

$$Y(s) = H(s)X(s),$$

we get

$$X(s) = \frac{Y(s)}{H(s)} = \frac{2(s+5)}{(s+3)(s+5)} = \frac{2}{s+3}.$$

Taking the inverse Laplace transform,

$$x(t) = 2e^{-3t}u(t).$$

(b) $2e^{-3t}u(t)$

Q5.14.2 MCQ 2014 1M Ans: B ● ○ ○

Use the Initial Value Theorem:

$$x(0^+) = \lim_{s\to\infty} sX(s).$$

Given,

$$X(s) = \frac{3s+5}{s^2+10s+21}.$$

Therefore,

$$x(0^+) = \lim_{s\to\infty} \frac{s(3s+5)}{s^2+10s+21} = \lim_{s\to\infty} \frac{3+\frac{5}{s}}{1+\frac{10}{s}+\frac{21}{s^2}} = 3.$$

(b) 3

Q5.13.1 MCQ 2013 1M Ans: C ● ○ ○

For a continuous-time LTI system to be **causal and BIBO stable**,

- All poles must lie in the left half of the s-plane, i.e.,

$$\Re(s) < 0.$$

  Hence, statements **A** and **D** are true.

- System zeros do not affect stability and may lie anywhere in the s-plane. Hence, statement **B** is true.

- Statement **C** is false because stability requires poles to satisfy

$$\Re(s) < 0,$$

not

$$|s| < 1.$$

For example, a pole at $s = -2$ is stable but lies outside the circle $|s| = 1$.

(c) All the poles must lie within $|s| = 1$

Q5.13.2 MCQ 2013 1M Ans: C ● ○ ○

Using Laplace transforms,

$$H(s) = \mathcal{L}\{tu(t)\} = \frac{1}{s^2},$$

and

$$X(s) = \mathcal{L}\{u(t-1)\} = \frac{e^{-s}}{s}.$$

Therefore,

$$Y(s) = H(s)X(s) = \frac{e^{-s}}{s^3}.$$

Taking the inverse Laplace transform,

$$y(t) = \frac{(t-1)^2}{2}u(t-1).$$

(c) $\frac{(t-1)^2}{2}u(t-1)$

Q5.12.1 MCQ 2012 1M Ans: D ● ○ ○

Given

$$F(s) = \mathcal{L}\{f(t)\} = \frac{1}{s^2+s+1}.$$

Using the differentiation property of the unilateral Laplace transform,

$$\mathcal{L}\{tf(t)\} = -\frac{dF(s)}{ds}.$$

Differentiating,

$$\frac{d}{ds}\left(\frac{1}{s^2+s+1}\right) = -\frac{2s+1}{(s^2+s+1)^2}.$$

Hence,

$$\mathcal{L}\{tf(t)\} = -\left(-\frac{2s+1}{(s^2+s+1)^2}\right) = \frac{2s+1}{(s^2+s+1)^2}.$$

(d) $\frac{2s+1}{(s^2+s+1)^2}$

Key Points

- Time multiplication property:

$$\mathcal{L}\{tf(t)\} = -\frac{dF(s)}{ds}.$$

- Differentiate the complete transform before applying the negative sign.
- This property is valid for both unilateral and bilateral Laplace transforms when the transform exists.

Q5.11.1 MCQ 2011 2M Ans: B ●○○

Let

$$F_1(s) = \mathcal{L}\{f(t)\}.$$

Since $F_2(s)$ is the Laplace transform of the delayed signal $f(t-\tau)$,

$$F_2(s) = e^{-s\tau}F_1(s).$$

Also,

$$|F_1(s)|^2 = F_1(s)F_1^*(s).$$

Hence,

$$G(s) = \frac{F_2(s)\,F_1^*(s)}{|F_1(s)|^2} = \frac{e^{-s\tau}F_1(s)F_1^*(s)}{F_1(s)F_1^*(s)} = e^{-s\tau}.$$

Using the time-delay property,

$$\mathcal{L}^{-1}\{e^{-s\tau}\} = \delta(t-\tau).$$

(b) An ideal delayed impulse $\delta(t-\tau)$

Q5.11.2 MCQ 2011 1M Ans: A ●○○

Since the signals exist for $t > 0$,

$$x(t) = e^{-t}u(t), \qquad y(t) = e^{-2t}u(t).$$

Taking Laplace transforms,

$$X(s) = \frac{1}{s+1}, \qquad Y(s) = \frac{1}{s+2}.$$

Using the convolution property,

$$Z(s) = X(s)Y(s) = \frac{1}{(s+1)(s+2)}.$$

Applying partial fractions,

$$\frac{1}{(s+1)(s+2)} = \frac{1}{s+1} - \frac{1}{s+2}.$$

Taking the inverse Laplace transform,

$$z(t) = e^{-t}u(t) - e^{-2t}u(t).$$

(a) $e^{-t} - e^{-2t}$

Q5.10.1 MCQ 2010 2M Ans: D ●○○

From the figure,

$$f(t) = \begin{cases} 1, & 0 \le t \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

The signal $g(t)$ is a unit-amplitude pulse over

$$3 \le t \le 5.$$

Assume

$$g(t) = f(at + b).$$

Since the pulse width changes from 1 to 2,

$$a = \frac{1}{2}.$$

Now,

$$\frac{t}{2} - \frac{3}{2} = 0 \quad \Rightarrow \quad t = 3,$$

and

$$\frac{t}{2} - \frac{3}{2} = 1 \quad \Rightarrow \quad t = 5.$$

Hence,

$$g(t) = f\left(\frac{t}{2} - \frac{3}{2}\right).$$

$$\boxed{\text{(d) } g(t) = f\left(\frac{t}{2} - \frac{3}{2}\right)}$$

Q5.10.2 MCQ 2010 2M Ans: C ● ○ ○

From the figure,

$$g(t) = u(t-3) - u(t-5).$$

Taking the Laplace transform,

$$G(s) = \frac{e^{-3s}}{s} - \frac{e^{-5s}}{s}.$$

Therefore,

$$G(s) = \frac{e^{-3s}}{s}\left(1 - e^{-2s}\right).$$

$$\boxed{\text{(c) } \frac{e^{-3s}}{s}\left(1 - e^{-2s}\right)}$$

Q5.07.1 MCQ 2007 2M Ans: C ● ○ ○

Using the time-shifting property of convolution,

$$x(t - t_1) * y(t - t_2) = (x * y)(t - t_1 - t_2).$$

Since

$$u(t+1) = u(t-(-1)), \qquad r(t-2) = r(t-2),$$

we obtain

$$u(t+1) * r(t-2) = (u * r)(t-1).$$

Now,

$$r(t) = tu(t).$$

Hence,

$$u(t) * r(t) = \int_{-\infty}^{\infty} u(\tau)\, r(t-\tau)\, d\tau.$$

Since

$$u(\tau) = 1, \quad \tau \geq 0,$$

and

$$r(t-\tau) = (t-\tau)u(t-\tau),$$

the limits of integration become

$$0 \le \tau \le t.$$

Therefore,

$$\begin{aligned} u(t) * r(t) &= \int_0^t (t-\tau)\, d\tau \\ &= \left[t\tau - \frac{\tau^2}{2}\right]_0^t \\ &= \frac{t^2}{2}. \end{aligned}$$

Including the causality condition,

$$u(t) * r(t) = \frac{t^2}{2}u(t).$$

Applying the shift $t \to t-1$,

$$u(t+1) * r(t-2) = \frac{1}{2}(t-1)^2 u(t-1).$$

$$\boxed{\text{(c) } \frac{1}{2}(t-1)^2 u(t-1)}$$

Q5.06.1 MCQ | 2006 | 2M | Ans: D | ● ○ ○

The running integrator is

$$y(t) = \int_{-\infty}^t x(t')\, dt'.$$

Differentiating,

$$\frac{dy(t)}{dt} = x(t).$$

Taking the bilateral Laplace transform,

$$sY(s) = X(s),$$

assuming the boundary term vanishes. Hence,

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s}.$$

Now examine the options:

- **A:** False. $H(s) = 1/s$ has a finite pole at $s = 0$.
- **B:** False. For a causal bounded input such as $u(t)$, the output is $tu(t)$, which is unbounded.
- **C:** True. An anti-causal bounded input exists only over $t \le 0$, so the accumulated integral remains finite, giving a bounded output.
- **D:** True. The transfer function $H(s) = 1/s$ has no finite zeros.

Since the question expects a single correct option, the property emphasized is that the running integrator has no finite zeros.

$$\boxed{\text{(d) Has no finite zeros in its double sided Laplace transform } Y(s).}$$

Q5.05.1 MCQ 2005 2M Ans: A ● ○ ○

To determine

$$\lim_{t\to\infty} f(t),$$

use the Final Value Theorem:

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s),$$

provided all poles of $sF(s)$ lie in the left half-plane.
Given,

$$F(s) = \frac{5s^2 + 23s + 6}{s(s^2 + 2s + 2)}.$$

Therefore,

$$sF(s) = \frac{5s^2 + 23s + 6}{s^2 + 2s + 2}.$$

Evaluating the limit,

$$\lim_{s\to 0} sF(s) = \frac{6}{2} = 3.$$

Since the poles of

$$s^2 + 2s + 2 = 0$$

are

$$s = -1 \pm j,$$

which lie in the left half-plane, the Final Value Theorem is applicable.

(a) 3

Q5.02.1 MCQ 2002 1M Ans: C ● ○ ○

The Final Value Theorem states that

$$\lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s),$$

provided all poles of $sY(s)$ lie in the left half-plane (except possibly a simple pole at the origin).
Hence, the final value of the function is obtained by evaluating

$$\lim_{s\to 0} sY(s).$$

(c) $\lim_{s\to 0} sY(s)$

Q5.99.1 MCQ 1999 2M Ans: D ● ○ ○

A rectangular current pulse of magnitude I and duration T is represented as

$$i(t) = I\,[u(t) - u(t - T)].$$

Taking the Laplace transform,

$$I(s) = I\left(\frac{1}{s} - \frac{e^{-Ts}}{s}\right).$$

Hence,

$$I(s) = \frac{I}{s}\left(1 - e^{-Ts}\right).$$

(d) $\frac{I}{s}\left(1 - e^{-Ts}\right)$

Q5.95.1 NAT | 1995 | 1M | Ans: * | ● ○ ○

$$f_1(t) = e^{-2t}u(t), \qquad f_2(t) = e^{-t}u(t).$$

Using the convolution property in the Laplace domain,

$$F_1(s) = \frac{1}{s+2}, \qquad F_2(s) = \frac{1}{s+1}.$$

Hence,

$$Y(s) = F_1(s)F_2(s) = \frac{1}{(s+1)(s+2)}.$$

Using partial fractions,

$$\frac{1}{(s+1)(s+2)} = \frac{1}{s+1} - \frac{1}{s+2}.$$

Taking the inverse Laplace transform,

$$f_1(t) * f_2(t) = \left(e^{-t} - e^{-2t}\right)u(t).$$

Therefore,

$$\boxed{\left(e^{-t} - e^{-2t}\right)u(t)}$$

PrepFusion

SOLUTIONS

VI

Z Transform, DTFS and DTFT

Topics

1. Introduction to Z-Transform
2. Z-Transform and ROC
3. Inverse Z-Transform
4. Important Properties of Z-Transform
5. Unilateral Z-Transform

Unit Snapshot*

Stream: EE	**Questions:** 24	**MCQ:** 23	**MSQ:** 0	**NAT:** 1
1M: 5	**2M:** 19	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–3M(2023)

**See preface for how Avg Weightage and Range are calculated.*

Q6.26.1 MCQ 2026 1M Ans: C ● ○ ○

For the given sequence,

$$x[n] = 0.9^{|n|}$$

Split it into right-sided and left-sided parts:

$$x[n] = 0.9^{n}u[n] + 0.9^{-n}u[-n-1].$$

The Z-transform of the right-sided sequence is

$$X_1(z) = \sum_{n=0}^{\infty}(0.9z^{-1})^{n},$$

which converges for

$$|z| > 0.9.$$

The Z-transform of the left-sided sequence is

$$X_2(z) = \sum_{n=-\infty}^{-1}(0.9^{-1}z)^{-n},$$

which converges for

$$|z| < \frac{1}{0.9}.$$

Hence, the overall ROC is the intersection of the two ROCs:

$$0.9 < |z| < \frac{1}{0.9}.$$

$$\boxed{\textbf{(c)}\ 0.9 < |z| < \frac{1}{0.9}}$$

Q6.24.1 MCQ 2024 2M Ans: C ● ○ ○

Method 1

Let

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}.$$

For

$$y[n] = x[2n],$$

its Z-transform is

$$Y(z) = \sum_{n=-\infty}^{\infty} x[2n]z^{-n}.$$

Using the even-index decomposition identity,

$$x[2n] = \frac{1}{2}\left(x_e[n] + x_o[n]\right),$$

which leads to the standard Z-transform relation

$$Y(z) = \frac{1}{2}\left[X\left(z^{1/2}\right) + X\left(-z^{1/2}\right)\right].$$

Method 2

The downsampling-by-2 property of the Z-transform states that if

$$y[n] = x[Mn],$$

then

$$Y(z) = \frac{1}{M}\sum_{k=0}^{M-1} X\left(W_M^k z^{1/M}\right),$$

where

$$W_M = e^{-j2\pi/M}.$$

For $M = 2$,

$$W_2 = -1,$$

hence

$$Y(z) = \frac{1}{2}\left[X\left(z^{1/2}\right) + X\left(-z^{1/2}\right)\right].$$

(c) $Y(z) = \frac{1}{2}\left[X\left(z^{1/2}\right) + X\left(-z^{1/2}\right)\right]$

Key Points

- Downsampling by M introduces M aliased copies in the Z-domain.

Q6.23.1 MCQ 2023 1M Ans: B ● ○ ○

The discrete-time Fourier transform (DTFT) exists only if the unit circle

$$|z| = 1$$

lies in the region of convergence (ROC) of the Z-transform.
The poles of

$$X(z) = \frac{4z}{\left(z - \frac{1}{5}\right)\left(z - \frac{2}{3}\right)(z - 3)}$$

are at

$$z = \frac{1}{5}, \quad \frac{2}{3}, \quad 3.$$

Possible ROCs are

$$|z| > 3, \qquad \frac{2}{3} < |z| < 3, \qquad \frac{1}{5} < |z| < \frac{2}{3}, \qquad |z| < \frac{1}{5}.$$

Among these, only

$$\frac{2}{3} < |z| < 3$$

contains the unit circle ($|z| = 1$). Hence, the DTFT exists only for this ROC.

(b) DTFT of $x[n]$ converges if R is $\frac{2}{3} < |z| < 3$.

Q6.23.2 NAT | 2023 | 2M | Ans: 0.038 | ● ○ ○

First expand the given DTFT:

$$X(\Omega) = \tfrac{1}{2} + e^{-j\Omega} + \tfrac{1}{2}e^{-2j\Omega}.$$

Thus the aperiodic sequence has samples $x[0] = 1/2$, $x[1] = 1$, $x[2] = 1/2$, and is zero elsewhere. The stated period-5 extension repeats these samples with two added zeros per period.
Its discrete-time Fourier-series coefficient is the corresponding DFT value divided by 5:

$$a_3 = \frac{1}{5}X(6\pi/5).$$

Since the phase factor has unit magnitude,

$$|a_3| = \frac{1 + \cos(6\pi/5)}{5}$$

$$|a_3| \simeq 0.0381966.$$

To three decimal places the answer is $\boxed{0.038}$, inside the existing 0.037 to 0.039 interval. The shortcut does not treat the original aperiodic sequence itself as periodic; it uses the explicitly defined periodic extension.

Q6.22.1 MCQ | 2022 | 2M | Ans: B | ● ○ ○

Since the DTFS coefficients are periodic with period N,

$$a_k = a_{k+N}.$$

Given,

$$N = 3, \qquad a_{-3} = 2, \qquad a_4 = 1.$$

Hence,

$$a_{-3} = a_0 = 2, \qquad a_4 = a_1 = 1, \qquad a_2 = 0.$$

Therefore,

$$x[n] = \sum_{k=0}^{2} a_k e^{j\frac{2\pi}{3}kn} = 2 + e^{j\frac{2\pi}{3}n}.$$

Rewrite the constant term as

$$2 = 1 + 1.$$

Thus,

$$x[n] = 1 + \left(1 + e^{j\frac{2\pi}{3}n}\right).$$

Using

$$1 = e^{j\frac{\pi}{3}n}e^{-j\frac{\pi}{3}n},$$

we get

$$1 + e^{j\frac{2\pi}{3}n} = e^{j\frac{\pi}{3}n}\left(e^{-j\frac{\pi}{3}n} + e^{j\frac{\pi}{3}n}\right).$$

Applying

$$e^{j\theta} + e^{-j\theta} = 2\cos\theta,$$

yields

$$x[n] = 1 + 2e^{j\frac{\pi}{3}n}\cos\left(\frac{\pi}{3}n\right).$$

Since

$$\frac{\pi}{3} = \frac{2\pi}{6},$$

the signal becomes

$$x[n] = 1 + 2e^{j\left(\frac{2\pi}{6}n\right)}\cos\left(\frac{2\pi}{6}n\right).$$

$$\boxed{\text{(b) } 1 + 2e^{j\left(\frac{2\pi}{6}n\right)} \cos\left(\frac{2\pi}{6}n\right)}$$

Key Points

- DTFS coefficients satisfy $a_k = a_{k+N}$.
- Convert all given indices to the range $0 \le k \le N-1$.

Q6.21.1 MCQ 2021 2M Ans: B ● ○ ○

Using the standard Z-transform pair,

$$a^n u[n] \xleftrightarrow{\mathcal{Z}} \frac{z}{z-a}.$$

Differentiating both sides with respect to a,

$$na^{n-1}u[n] \xleftrightarrow{\mathcal{Z}} \frac{z}{(z-a)^2}.$$

Multiplying both sides by a,

$$na^n u[n] \xleftrightarrow{\mathcal{Z}} \frac{az}{(z-a)^2}.$$

Now,

$$\frac{z^2}{(z-a)^2} = \frac{z}{a} \cdot \frac{az}{(z-a)^2}.$$

Using the property

$$a^n u[n] \xleftrightarrow{\mathcal{Z}} \frac{z}{z-a}$$

we get

$$x[n] = \frac{1}{a}(n+1)a^{n+1}u[n] = (n+1)a^n u[n].$$

$$\boxed{\textbf{(b) } (n+1)a^n u[n]}$$

Q6.20.1 MCQ 2020 1M Ans: D ● ○ ○

Given

$$x[n] = \left(\frac{1}{2}\right)^n u[n].$$

Its Z-transform is

$$X(z) = \frac{1}{1-\frac{1}{2}z^{-1}}, \qquad \text{ROC: } |z| > \frac{1}{2}.$$

The signal is delayed by k samples:

$$y[n] = x[n-k], \qquad k > 0.$$

Using the time-shifting property,

$$x[n-k] \xleftrightarrow{\mathcal{Z}} z^{-k}X(z).$$

Hence,

$$Y(z) = \frac{z^{-k}}{1-\frac{1}{2}z^{-1}}.$$

Since a finite delay does not change the ROC,

$$\boxed{|z| > \frac{1}{2}}.$$

(d) $|z| > \frac{1}{2}$

Q6.17.1 MCQ | 2017 | 1M | Ans: B

Determine the transfer function corresponding to each pole-zero plot and evaluate its frequency response.

System P:

$$H(z) = \frac{z^2 - 1}{z^2} = 1 - z^{-2}.$$

Substituting $z = e^{j\omega}$,

$$H(e^{j\omega}) = 1 - e^{-j2\omega} = e^{-j\omega}\left(e^{j\omega} - e^{-j\omega}\right) = 2je^{-j\omega}\sin\omega.$$

Hence,

$$|H(e^{j\omega})| = 2|\sin\omega|.$$

Since

$$|H(e^{j0})| = 0, \qquad |H(e^{j\pi})| = 0,$$

the response is maximum around

$$\omega = \frac{\pi}{2},$$

so P is a band-pass filter.

System Q:

$$H(z) = \frac{z^2 - 1}{z^2 + 0.25}.$$

Therefore,

$$H(e^{j\omega}) = \frac{e^{j2\omega} - 1}{e^{j2\omega} + 0.25}.$$

The numerator becomes zero at

$$\omega = 0,\ \pi,$$

so the response is zero at both ends of the spectrum. Hence, Q is also a band-pass filter.

System R:

$$H(z) = \frac{z^2 - 1}{z^2 + 1}.$$

Thus,

$$H(e^{j\omega}) = \frac{e^{j2\omega} - 1}{e^{j2\omega} + 1}.$$

Again,

$$e^{j2\omega} - 1 = 0$$

at

$$\omega = 0,\ \pi,$$

so the response is maximum only at intermediate frequencies. Hence, R is also a band-pass filter.

(b) All three are band-pass filters.

Key Points

- A zero at $z = 1$ suppresses the DC component ($\omega = 0$).
- A zero at $z = -1$ suppresses the highest frequency component ($\omega = \pi$).
- A filter with zero response at both $\omega = 0$ and $\omega = \pi$ behaves as a band-pass filter.

Mistakes to be avoided

- Do not classify the filter only from the pole locations.
- Always verify the frequency response by substituting $z = e^{j\omega}$.
- Check the response at $\omega = 0$ and $\omega = \pi$ before deciding the filter type.

Q6.17.2 MCQ 2017 2M Ans: D ●○○

The equivalent impulse response of the cascaded system is the convolution of the individual impulse responses.

$$h(n) = h_1(n) * h_2(n)$$

Given,

$$h_1(n) = \{1, -1\}, \qquad h_2(n) = \{1, 1\}$$

Therefore,

$$h(n) = \{1, -1\} * \{1, 1\} = \{1, 0, -1\}$$

Hence,

$$y(n) = x(n) * h(n)$$

Checking the options, for

$$x(n) = \{1, 2, 2, 1\}$$

we get

$$y(n) = \{1, 2, 2, 1\} * \{1, 0, -1\} = \{1, 2, 1, -1, -2, -1\}$$

which matches the given output sequence.

(d) $x(n) = \{1, 2, 2, 1\}$

Q6.15.1 MCQ 2015 2M Ans: C ●○○

Let

$$x[n] = (-0.25)^n u[n] + (0.5)^n u[-n-1].$$

Find the ROC of each sequence separately.
For the right-sided sequence,

$$(-0.25)^n u[n]$$

the ROC is

$$|z| > 0.25.$$

For the left-sided sequence,

$$(0.5)^n u[-n-1],$$

the ROC is

$$|z| < 0.5.$$

Since the given signal is a sum of these two sequences, the overall ROC is the intersection of the individual ROCs:

$$|z| > 0.25 \quad \text{and} \quad |z| < 0.5.$$

Hence,

$$0.25 < |z| < 0.5.$$

(c) The annular region between the two circles centered at the origin with radii 0.25 and 0.5.

Q6.15.2 MCQ 2015 2M Ans: A ● ○ ○

The first difference of a sequence is

$$y[n] = x[n] - x[n-1].$$

Using the time-shifting property,

$$x[n-1] \xleftrightarrow{Z} z^{-1}X(z).$$

Hence,

$$Y(z) = X(z) - z^{-1}X(z) = (1 - z^{-1})X(z).$$

Given

$$X(z) = 2z + 4 - \frac{4}{z} + \frac{3}{z^2},$$

we get

$$Y(z) = \left(1 - \frac{1}{z}\right)\left(2z + 4 - \frac{4}{z} + \frac{3}{z^2}\right).$$

Expanding,

$$Y(z) = 2z + 4 - \frac{4}{z} + \frac{3}{z^2} - 2 - \frac{4}{z} + \frac{4}{z^2} - \frac{3}{z^3}.$$

Therefore,

$$Y(z) = 2z + 2 - \frac{8}{z} + \frac{7}{z^2} - \frac{3}{z^3}.$$

(a) $Y(z) = 2z + 2 - \frac{8}{z} + \frac{7}{z^2} - \frac{3}{z^3}.$

Q6.14.1 MCQ 2014 2M Ans: B ● ○ ○

Given,

$$X(z) = \frac{1}{1 - z^{-3}}.$$

Using the geometric series expansion,

$$\frac{1}{1-r} = 1 + r + r^2 + r^3 + \cdots, \qquad |r| < 1.$$

Here,

$$r = z^{-3},$$

so

$$X(z) = 1 + z^{-3} + z^{-6} + z^{-9} + \cdots.$$

Using the inverse Z-transform,

$$x[n] = \delta[n] + \delta[n-3] + \delta[n-6] + \cdots.$$

Hence,

$$x[2] = 0, \qquad x[3] = 1.$$

(b) $x[2] = 0,\ x[3] = 1.$

Q6.14.2 MCQ 2014 2M Ans: C ● ○ ○

The given input signal is

$$x(t) = 2 + 5\sin(100\pi t).$$

Since

$$100\pi = 2\pi f,$$

the sinusoidal frequency is

$$f = \frac{100\pi}{2\pi} = 50\,\text{Hz}.$$

The sampling frequency is

$$f_s = 400\,\text{Hz}.$$

Hence, the number of samples per cycle is

$$N = \frac{f_s}{f} = \frac{400}{50} = 8.$$

The given system is

$$\frac{Y(z)}{X(z)} = \frac{1}{N}\left(\frac{1 - z^{-N}}{1 - z^{-1}}\right),$$

which is an N-point moving average filter.
Over one complete cycle of the sampled sinusoid,

$$\frac{1}{N}\sum_{k=0}^{N-1} \sin\left(\frac{2\pi k}{N}\right) = 0.$$

The DC component remains unchanged since

$$\frac{1}{N}\sum_{k=0}^{N-1} 2 = 2.$$

Therefore, under steady state,

$$y[n] = 2.$$

(c) 2

Q6.12.1 MCQ 2012 1M Ans: C ● ○ ○

Given,

$$x[n] = \left(\frac{1}{3}\right)^{|n|} - \left(\frac{1}{2}\right)^{n} u[n].$$

Express the first term as

$$\left(\frac{1}{3}\right)^{|n|} = \left(\frac{1}{3}\right)^{n} u[n] + 3^{n} u[-n-1].$$

Hence,

$$x[n] = \left[\left(\frac{1}{3}\right)^{n} - \left(\frac{1}{2}\right)^{n}\right] u[n] + 3^{n} u[-n-1].$$

Now determine the ROC of each part.
For

$$\left(\frac{1}{3}\right)^{n} u[n],$$

$$\left(\frac{1}{3}\right)^{n} u[n] \xleftrightarrow{\mathcal{Z}} \frac{1}{1 - \frac{1}{3}z^{-1}}, \qquad |z| > \frac{1}{3}.$$

For

$$\left(\frac{1}{2}\right)^n u[n],$$

$$\left(\frac{1}{2}\right)^n u[n] \xleftrightarrow{\mathcal{Z}} \frac{1}{1-\frac{1}{2}z^{-1}}, \qquad |z| > \frac{1}{2}.$$

Since these two right-sided sequences are added together, their ROC is

$$|z| > \frac{1}{2}.$$

For

$$3^n u[-n-1],$$

$$3^n u[-n-1] \xleftrightarrow{\mathcal{Z}} -\frac{1}{1-3z^{-1}}, \qquad |z| < 3.$$

Therefore, the overall ROC is the intersection:

$$|z| > \frac{1}{2} \quad \text{and} \quad |z| < 3.$$

Hence,

$$\frac{1}{2} < |z| < 3.$$

$$\boxed{\text{(c) } \frac{1}{2} < |z| < 3.}$$

Q6.09.1 MCQ | 2009 | 2M | Ans: A | ● ○ ○

Given,

$$X(z) = 4z^{-3} + 3z^{-1} + 2 - 6z^2 + 2z^3,$$

and

$$H(z) = 3z^{-1} - 2.$$

The output is

$$Y(z) = H(z)X(z).$$

Expanding,

$$\begin{aligned} Y(z) &= (3z^{-1} - 2)(4z^{-3} + 3z^{-1} + 2 - 6z^2 + 2z^3) \\ &= -4z^3 + 18z^2 - 18z + 12z^{-4} + 9z^{-2} - 4 - 8z^{-3}. \end{aligned}$$

Since

$$Y(z) = \sum_{n=-\infty}^{\infty} y[n]z^{-n},$$

positive powers of z correspond to samples with $n < 0$.
As $Y(z)$ contains positive powers $z,\ z^2,\ z^3$, the sequence $y[n]$ is non-causal.
Also, $Y(z)$ contains only finitely many powers of z, so $y[n]$ has finite support.
Hence, the output is a non-causal finite-duration sequence.

$$\boxed{\text{(a) } y[n] \text{ is non-causal with finite support.}}$$

Q6.08.1 MCQ 2008 2M Ans: D ● ○ ○

The inverse Z-transform using the residue method is

$$x[n] = \sum \text{Res}\left\{X(z)z^{n-1}\right\}.$$

Given,

$$X(z) = \frac{z}{(z-a)^2}.$$

Hence,

$$X(z)z^{n-1} = \frac{z^n}{(z-a)^2}.$$

Since $z = a$ is a double pole,

$$\text{Res}\left\{\frac{z^n}{(z-a)^2}\right\}_{z=a} = \frac{d}{dz}(z^n)\Big|_{z=a}.$$

Now,

$$\frac{d}{dz}(z^n) = nz^{n-1}.$$

Therefore,

$$\text{Res}\left\{X(z)z^{n-1}\right\}_{z=a} = na^{n-1}.$$

(d) na^{n-1}.

Q6.08.2 MCQ 2008 2M Ans: B ● ○ ○

Given,

$$x[n] = (1+j)^n.$$

For an LTI system, an exponential input is an eigenfunction. Hence,

$$y[n] = H(1+j)(1+j)^n.$$

Since the output is zero,

$$H(1+j) = 0.$$

As the system is real,

$$H(1-j) = 0$$

also holds.
Thus, $H(z)$ must contain at least two zeros at

$$z = 1+j, \qquad z = 1-j.$$

Further, the ROC of

$$\left(1 - \frac{1}{2}z^{-1}\right)H(z)$$

is the entire z-plane (except $z = 0$). Hence, the factor

$$\left(1 - \frac{1}{2}z^{-1}\right)$$

must cancel a pole of $H(z)$ at

$$z = \frac{1}{2}.$$

Therefore, $H(z)$ has a minimum of one pole and two zeros.

(b) One pole and two zeros.

Q6.07.1 MCQ 2007 2M Ans: A ● ○ ○

The given FIR filter is

$$G(z) = \alpha z^{-1} + \beta z^{-3}.$$

Its impulse response is

$$h[n] = \{0, \alpha, 0, \beta\}.$$

For an FIR filter to have linear phase, the impulse response must be symmetric or anti-symmetric.
Symmetric condition:

$$h[1] = h[3] \quad \Rightarrow \quad \alpha = \beta.$$

Anti-symmetric condition:

$$h[1] = -h[3] \quad \Rightarrow \quad \alpha = -\beta.$$

Since the filter is specified to be a **low-pass** filter, the DC gain must be non-zero.
Now,

$$G(e^{j0}) = \alpha + \beta.$$

If

$$\alpha = -\beta,$$

then

$$G(e^{j0}) = 0,$$

which cannot represent a low-pass filter.
Hence, the valid condition is

$$\alpha = \beta.$$

(a) $\alpha = \beta$.

Key Points

- Linear-phase FIR filters require symmetric or anti-symmetric impulse responses.
- Low-pass FIR filters require a symmetric impulse response.
- Anti-symmetric FIR filters have zero DC gain.

Q6.07.2 MCQ 2007 2M Ans: C ● ○ ○

For distortion-free transmission, the output must be a delayed and scaled version of the input:

$$y[n] = Kx[n - n_0].$$

Taking the Z-transform,

$$Y(z) = Kz^{-n_0}X(z).$$

Hence, the transfer function is

$$G(z) = Kz^{-n_0}.$$

Its frequency response is

$$G(e^{j\omega}) = Ke^{-j\omega n_0}.$$

Therefore, the phase response is

$$\angle G(e^{j\omega}) = -\omega n_0,$$

which is a linear function of frequency.

(c) Provide linear phase shift that is proportional to frequency.

Q6.07.3 MCQ 2007 2M Ans: B ● ○ ○

Given,

$$X(z) = 1 - 3z^{-1}, \qquad Y(z) = 1 + 2z^{-2}.$$

Also,

$$h[n] = x[n-1] * y[n].$$

Using the time-shifting property,

$$x[n-1] \xleftrightarrow{\mathcal{Z}} z^{-1}X(z).$$

Since convolution in the time domain corresponds to multiplication in the Z-domain,

$$H(z) = z^{-1}X(z)Y(z).$$

For the input

$$\delta[n-1],$$

$$\delta[n-1] \xleftrightarrow{\mathcal{Z}} z^{-1}.$$

Hence,

$$Z(z) = H(z)\,z^{-1} = z^{-2}X(z)Y(z).$$

Now,

$$X(z)Y(z) = (1 - 3z^{-1})(1 + 2z^{-2}) = 1 - 3z^{-1} + 2z^{-2} - 6z^{-3}.$$

Therefore,

$$Z(z) = z^{-2} - 3z^{-3} + 2z^{-4} - 6z^{-5}.$$

Taking the inverse Z-transform,

$$z[n] = \delta[n-2] - 3\delta[n-3] + 2\delta[n-4] - 6\delta[n-5].$$

(b) $\delta[n-2] - 3\delta[n-3] + 2\delta[n-4] - 6\delta[n-5]$.

Q6.07.4 MCQ 2007 2M Ans: A ● ○ ○

Given,

$$g[0] = 0, \qquad g[1] = g[2] = 1.$$

Hence,

$$G(z) = z^{-1} + z^{-2}.$$

Since the feedback is positive,

$$T(z) = \frac{G(z)}{1 - KG(z)} = \frac{z^{-1} + z^{-2}}{1 - K(z^{-1} + z^{-2})}.$$

The characteristic equation is

$$1 - Kz^{-1} - Kz^{-2} = 0.$$

Multiplying by z^2,

$$z^2 - Kz - K = 0.$$

For the polynomial

$$z^2 + a_1 z + a_2,$$

the Jury stability conditions are

$$1 + a_1 + a_2 > 0, \qquad 1 - a_1 + a_2 > 0, \qquad 1 - a_2 > 0.$$

Here,

$$a_1 = -K, \qquad a_2 = -K.$$

Therefore,

$$1 - 2K > 0 \;\Rightarrow\; K < \frac{1}{2},$$

$$1 + K - K = 1 > 0,$$

and

$$1 + K > 0 \;\Rightarrow\; K > -1.$$

Hence,

$$-1 < K < \frac{1}{2}.$$

(a) $[-1, \frac{1}{2}]$.

Q6.06.1 MCQ | 2006 | 2M | Ans: D | ● ○ ○

Given,

$$y[n] - 0.8y[n-1] = x[n] + 1.25x[n+1].$$

Taking the Z-transform,

$$Y(z) - 0.8z^{-1}Y(z) = X(z) + 1.25zX(z).$$

Hence,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 + 1.25z}{1 - 0.8z^{-1}} = \frac{z(1 + 1.25z)}{z - 0.8}.$$

Rewrite $H(z)$ as

$$H(z) = 1.25\left[\frac{(z - 0.8) + 1.6}{z - 0.8}\right] = 1.25\left[1 + \frac{1.6}{z - 0.8}\right].$$

Using

$$\frac{1}{1 - 0.8z^{-1}} \overset{\mathcal{Z}}{\longleftrightarrow} (0.8)^n u[n],$$

we obtain

$$H(z) = 1.25z + \frac{2}{1 - 0.8z^{-1}}.$$

Taking the inverse Z-transform,

$$h[n] = 1.25\delta[n+1] + 2(0.8)^n u[n].$$

The term

$$1.25\delta[n+1]$$

makes the impulse response non-causal.
Also,

$$h[n] \geq 0, \qquad \forall n.$$

Hence, the right-sided impulse response is non-negative.

(d) Non-negative

Q6.06.2 MCQ 2006 2M Ans: B ● ○ ○

Two discrete-time signals are orthogonal if

$$\sum_{n=-\infty}^{\infty} x_1[n]x_2[n] = 0.$$

Given,

$$X(z) = \sum_{n=0}^{\infty} \frac{3^n}{n+2} z^{-2n}.$$

Hence,

$$x[n] \neq 0 \quad \text{only for even values of } n.$$

Therefore, an orthogonal sequence must have non-zero samples only at odd indices so that

$$x[n]y[n] = 0, \qquad \forall n.$$

Among the given options,

$$Y_2(z) = \sum_{n=0}^{\infty} (5^{-n}) z^{-(2n+1)}$$

contains only odd powers of z^{-1}. Thus,

$$y_2[n] \neq 0 \quad \text{only for odd values of } n.$$

Hence,

$$\sum_{n=-\infty}^{\infty} x[n]y_2[n] = 0.$$

Therefore, Option B is correct.

$$\boxed{\text{(b) } y_2[n] \leftrightarrow Y_2(z) = \sum_{n=0}^{\infty} (5^{-n})\, z^{-(2n+1)}}$$

Q6.05.1 MCQ 2005 2M Ans: B ● ○ ○

Given,

$$F(z) = \frac{1}{z+1} = \frac{z^{-1}}{1+z^{-1}} = \frac{z^{-1}}{1-(-1)z^{-1}}.$$

Using the standard pair,

$$\frac{z^{-1}}{1-az^{-1}} \xleftrightarrow{\mathcal{Z}} a^{n-1}u[n-1],$$

with

$$a = -1,$$

we obtain

$$f[k] = (-1)^{k-1}u[k-1].$$

Since

$$u[k-1] = u[k] - \delta[k],$$

we get

$$\begin{aligned} f[k] &= (-1)^{k-1}(u[k] - \delta[k]) \\ &= -(-1)^k u[k] + \delta[k]. \end{aligned}$$

For $k \geq 0$,

$$u[k] = 1.$$

Hence,

$$f[k] = \delta[k] - (-1)^k.$$

$$\boxed{\text{(b) } \delta(k) - (-1)^k.}$$

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